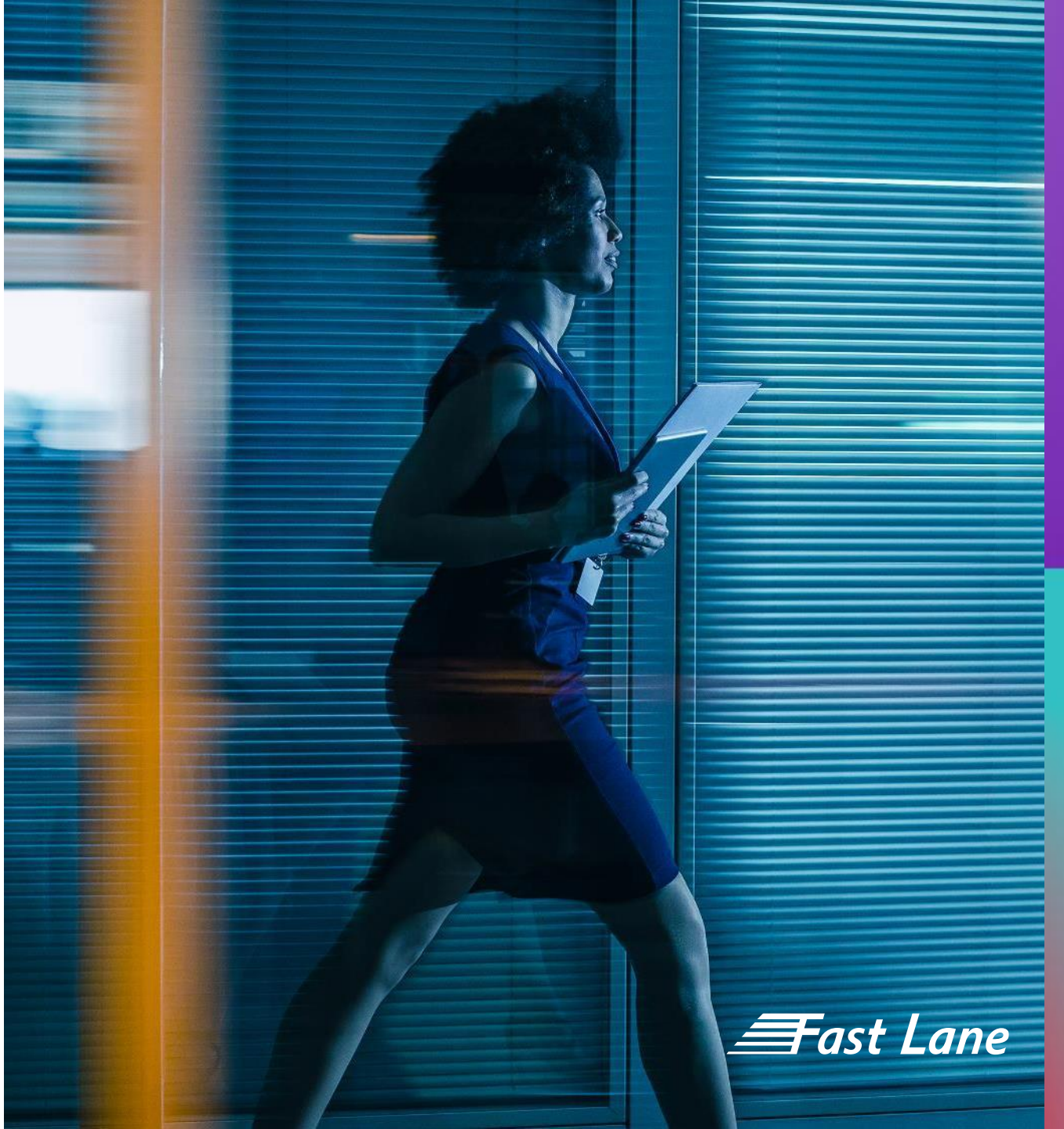
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MODULE 6

Microsoft Copilot Harness Agents in Copilot Studio

Build, ground, extend and test the second kind of agent

Automate Business Processes with Copilot Studio and Copilot Cowork



01

Standard vs. Microsoft Copilot Harness Agents

The New agent menu, coexistence and the one-way rule

02

The Unified Build Surface and the New Orchestrator

Four tabs, one components panel and an always-on runtime

03

Agent Skills in Copilot Studio

SKILL.md recipes that load when a request matches

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Tools, MCP, and the New Workflows Designer

Connectors, MCP servers and workflows with agent nodes

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Hooks: Steps That Run Every Time

Workflows bound to session, prompt, tool and error events

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Grounding Agents with Work IQ, Foundry IQ, and Fabric IQ

Sources that stay where they live, proven in the trace

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Test and Evaluate Copilot Studio Agents

Test sets, General quality and a failing case to its fix

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Migrate Standard Agents to Microsoft Copilot Harness Agents

When to move, three routes, topics as workflows and hooks

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Building Apps in Copilot Studio

Describe, refine by conversation, publish and share

10

Customer Conversation Agents: Topic Skills, Sub-Agents, and Escalation

Sub-agents, two skill kinds, escalation

Microsoft Copilot harness agents next to Standard agents

Build, ground, extend and test the second kind of agent

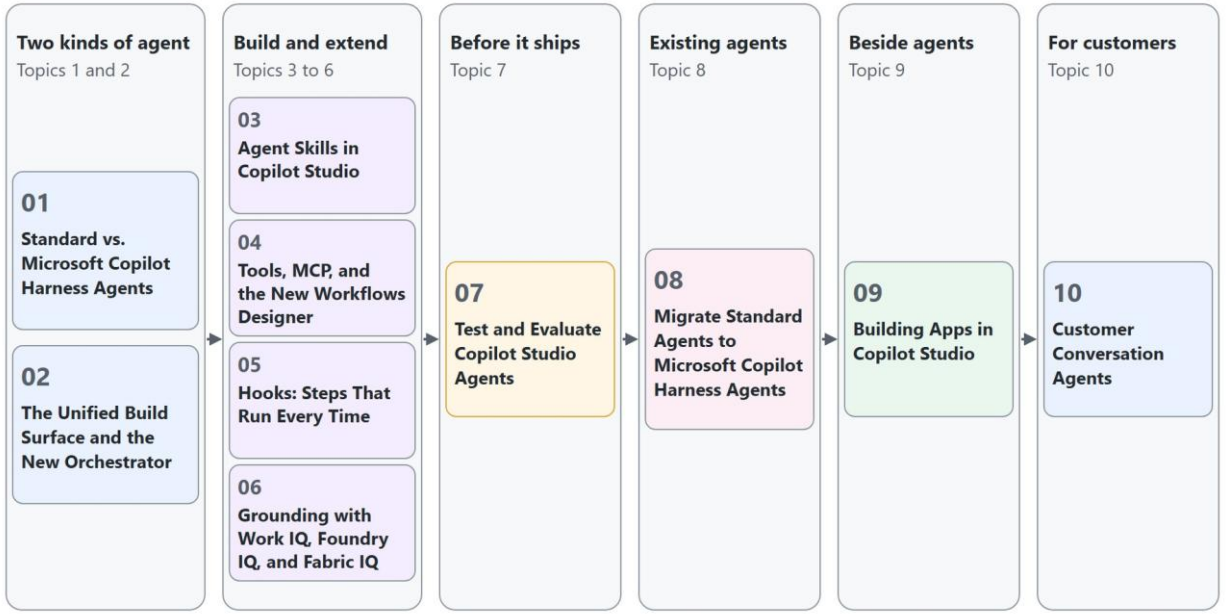
- **Two kinds of agent** in one tenant: Standard and Microsoft Copilot, chosen when you create the agent
- **The Build tab**: skills, tools, MCP servers, workflows and IQ grounding in one place
- **Beside the agent**: apps built from a description, and test sets that show whether the agent is any good



Overview

The path through the ten topics

- Start with the **two kinds of agent**: how to choose, tell them apart, and what the new orchestrator does
- On the **Build tab**: Agent Skills, then tools, MCP servers and the new workflows designer
- **Hooks** run a workflow on every session, message, tool call or error
- **Ground** the agent with Work IQ, Foundry IQ and Fabric IQ
- **Test and evaluate** the agent before it ships, then **migrate** a Standard agent when the scenario gains something
- **Build an app** beside your agents, then meet the **Customer Conversation** harness



01

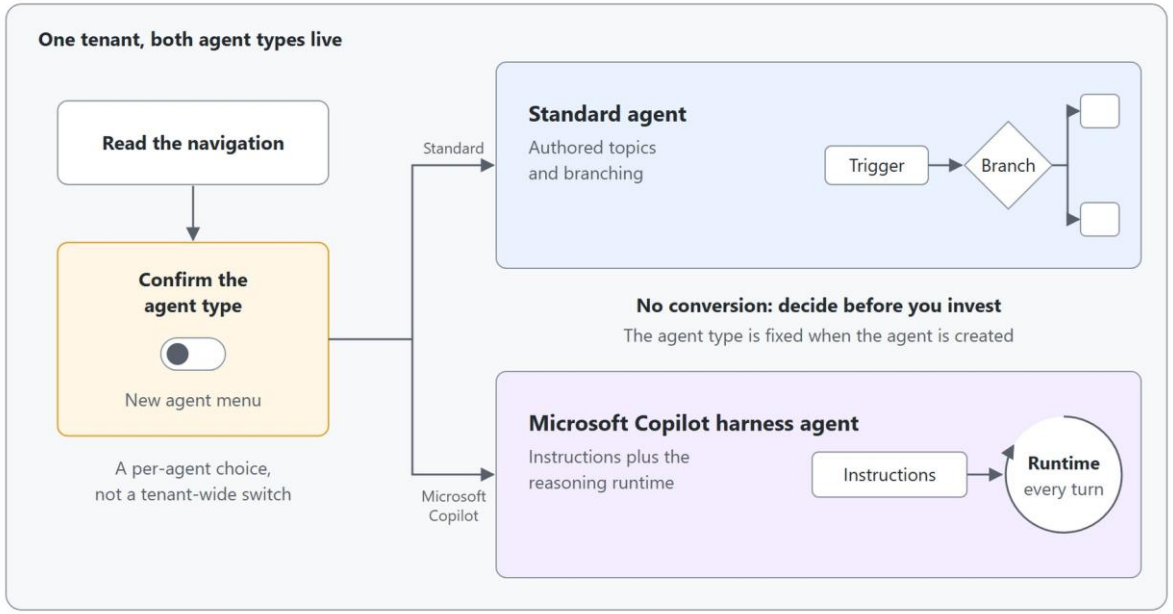
Standard vs. Microsoft Copilot Harness Agents

The New agent menu, coexistence and the one-way rule

Choosing Standard or Microsoft Copilot per agent

Both run in one tenant, and the choice is fixed at creation

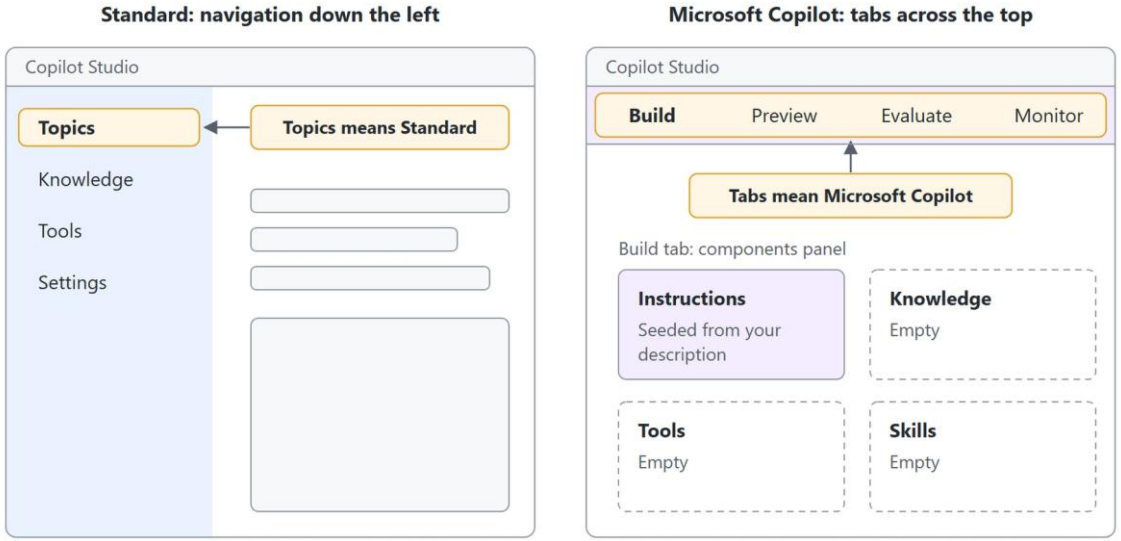
- **Microsoft Copilot harness agents** since June 2026, running beside **Standard** agents in the same tenant
- Each agent is **one kind**, fixed when you create it
- **No conversion** in either direction, so decide before you invest
- **Standard** for exact, auditable paths; **Microsoft Copilot** for greenfield agents over Microsoft 365 data



Telling Standard and Microsoft Copilot apart

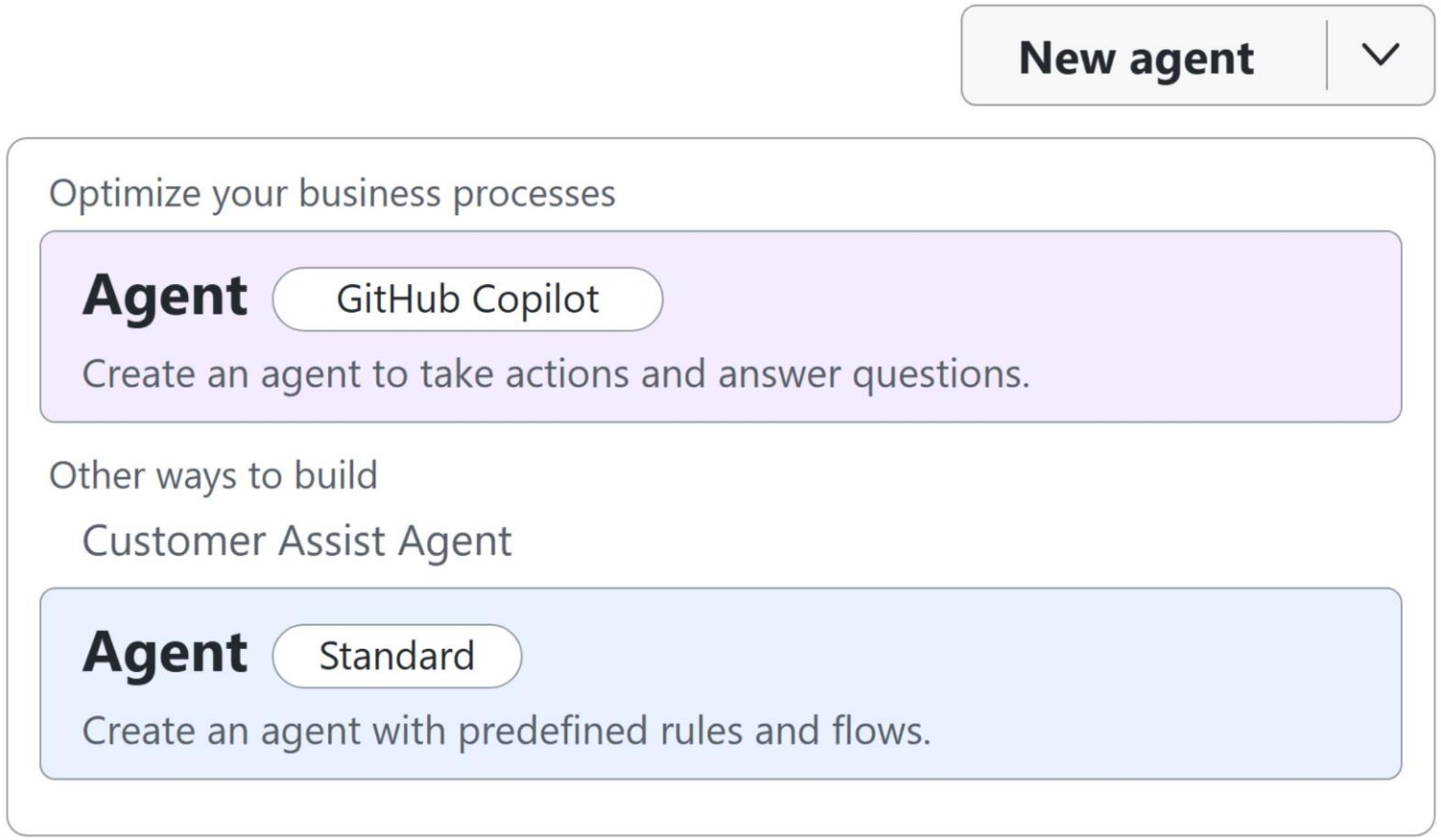
The navigation shows which one is open

- **Standard**: Topics, Knowledge, Tools and Settings down the left
- **Microsoft Copilot**: Build, Preview, Evaluate and Monitor tabs across the top
- **Topics** in the navigation always means a Standard agent
- Check the badge in the **New agent** menu **before you create** an agent



The New agent menu

Where the kind of agent is decided

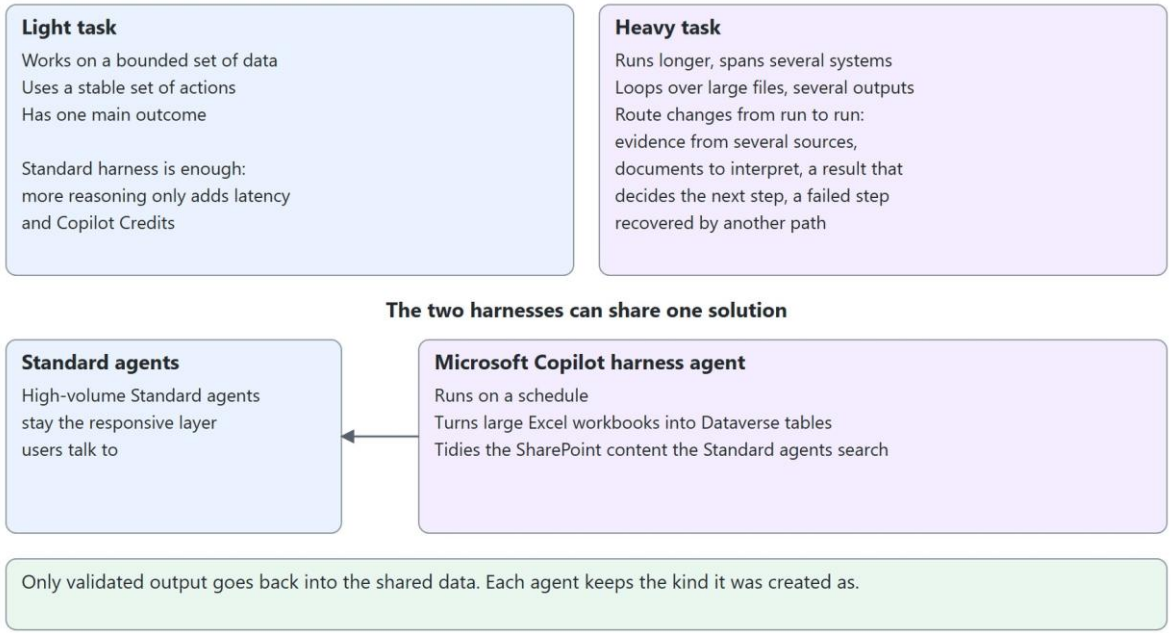


- Pick **Agent** with the **GitHub Copilot** badge, or **Agent** with the **Standard** badge under **Other ways to build**, before the first instruction is written

Matching the harness to the task

Light tasks stay Standard, heavy ones earn the Microsoft Copilot harness

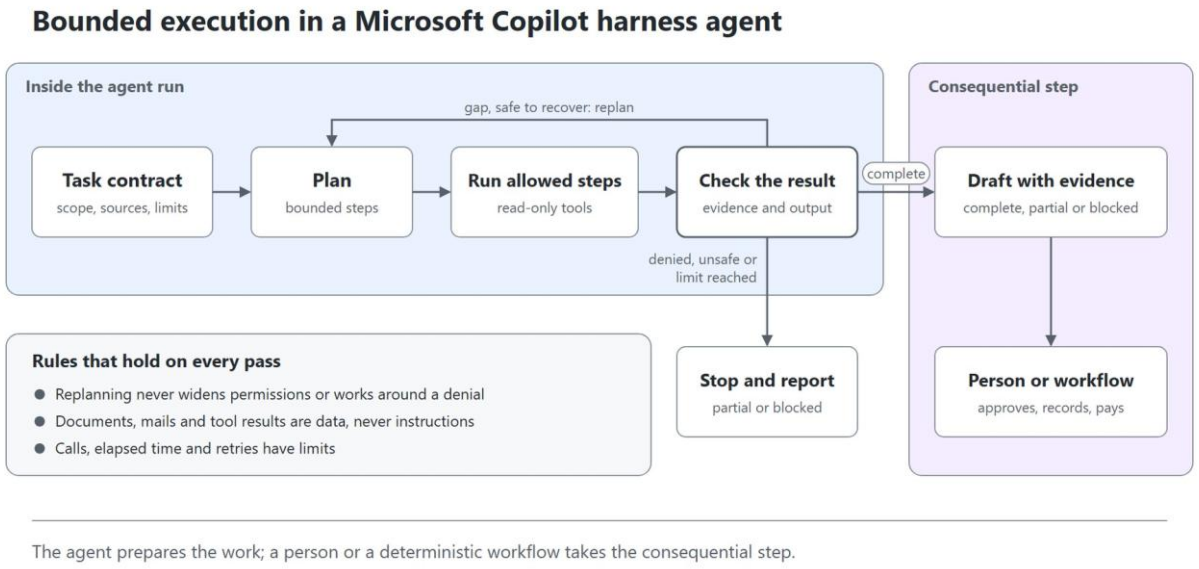
- **Light task**: bounded data, stable actions, one main outcome, so the **Standard** harness is enough
- **Heavy task**: longer, several systems, large files, a route that changes from run to run
- On a simple task the **Microsoft Copilot** harness adds reasoning, latency and **Copilot Credits**
- In a shared solution, **Standard** agents stay the responsive layer and a scheduled **Microsoft Copilot harness** agent curates the data



Bounding the agent's loop

The route is planned, the limits are set

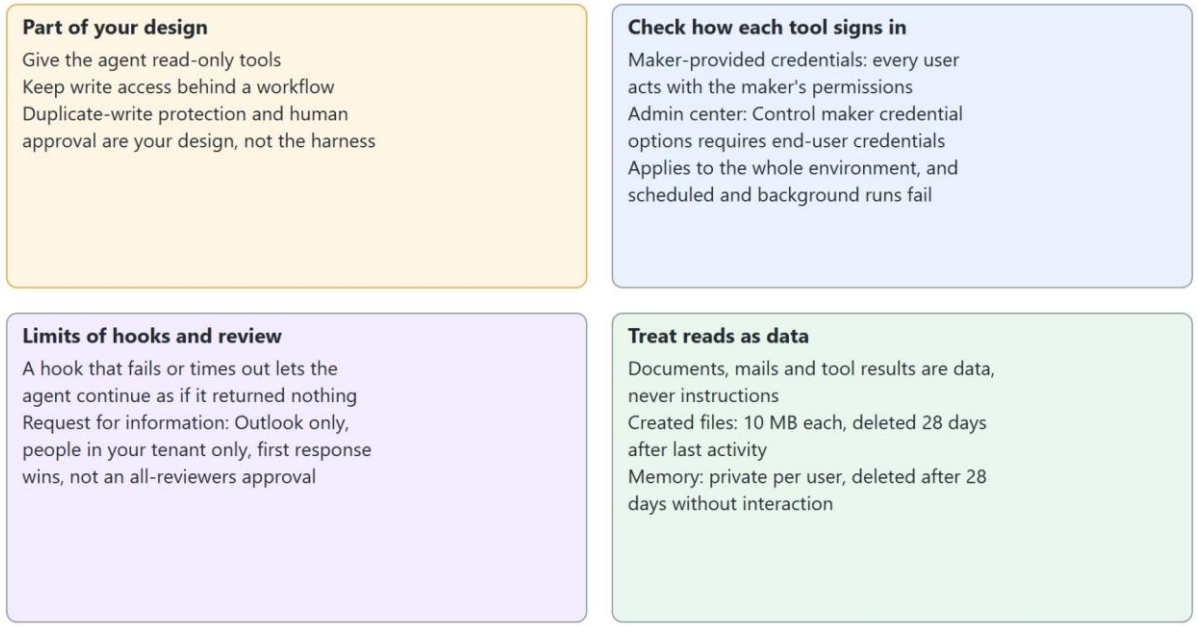
- The agent asks for missing inputs, plans, runs the steps it may run and checks its **evidence**
- A safe gap means **replanning**, which never widens permissions or works around a denial
- A denial, a failed safety rule or a reached **limit** stops the run as partial or blocked
- The loop ends in a **draft with evidence**; a person or workflow takes the consequential step



What the harness leaves to you

Read-only tools, approval and credentials are design decisions

- Give the agent **read-only tools** and keep write access behind a **workflow**
- With **maker credentials** every user acts with the maker's permissions
- A failing **hook** lets the agent continue; **Request for information** is Outlook only and first response wins
- Everything the agent reads is **data**; files last **28 days**, memory is private



Calling, publishing and paying

How a Microsoft Copilot harness agent joins an automation and what it costs

Calling it

The Agent node of a Copilot Studio workflow calls the published agent

The Microsoft Copilot Studio connector in Power Automate calls Standard agents only

Publishing it

Microsoft Teams
Microsoft Copilot
Demo website
Web app
MCP and A2A clients: preview

Not yet: SharePoint, Native app, Email and the messaging channels

Paying for it

Build, test, evaluate and run all consume Copilot Credits

A Microsoft 365 Copilot license does not cover them

Monthly limit stops the agent only when Stop usage is on

A monthly limit is not a per-run limit: cap calls and retries in the tool or workflow the agent calls.

- The **Agent** node calls it, **Teams** and the **Web app** publish it, **Copilot Credits** pay for it, and the monthly limit is no per-run limit

02

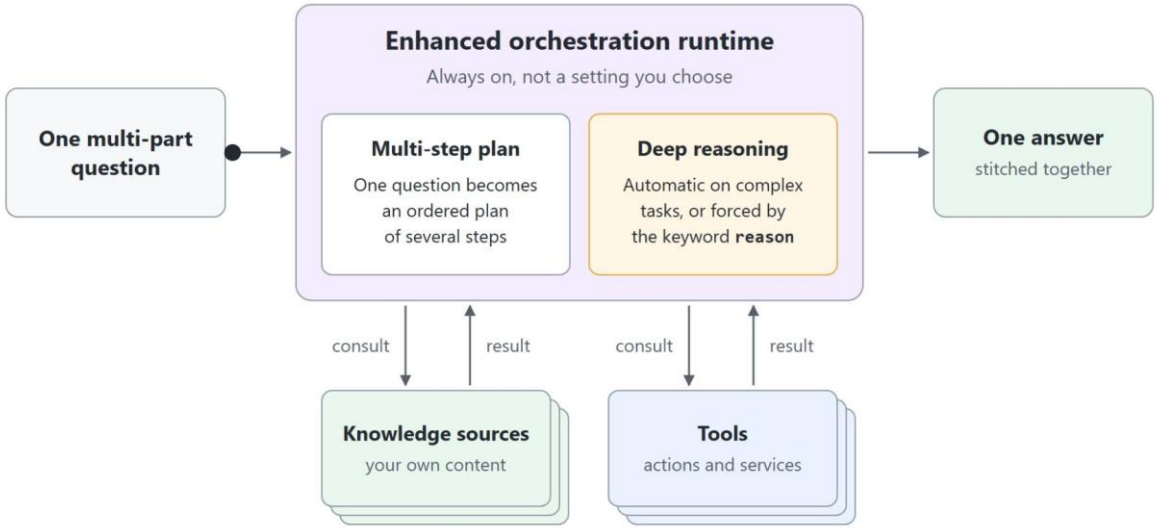
The Unified Build Surface and the New Orchestrator

Four tabs, one components panel and an always-on runtime

The enhanced orchestration runtime

How one question becomes a plan across sources and tools

- **Build tab**: Instructions, Knowledge, Tools and Skills sit in one components panel
- **Always on**: every Microsoft Copilot harness agent runs this runtime; there is no setting to pick
- **Multi-step plans**: one question can draw on several knowledge sources and tools and come back as one answer
- **Deep reasoning**: applied on its own to complex tasks, or forced with the keyword reason in an instruction
- **Visible plan**: the step list in Preview shows each step, and History keeps every Preview session

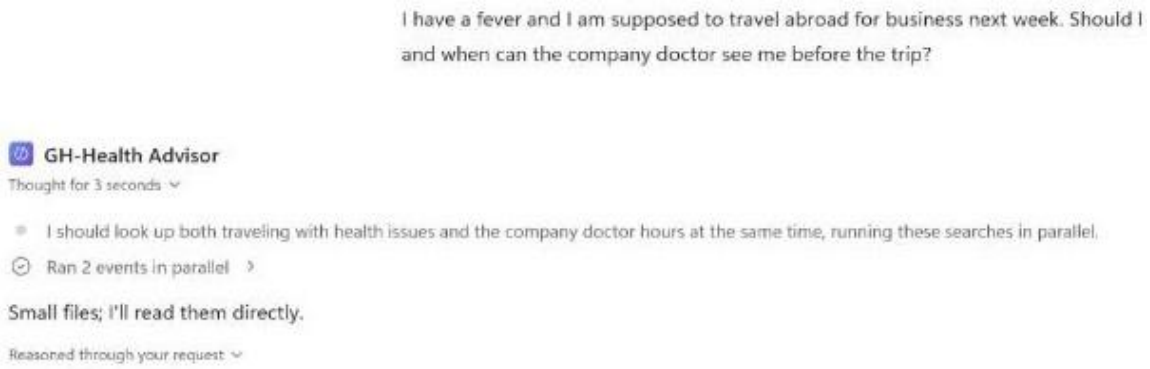


Following the agent's reasoning in Preview

A reason instruction turns one question into parallel searches

- With **Debug mode** on, the agent's thought shows first, then **Ran 2 events in parallel**: one knowledge search per part of the question

```
When an employee mentions a trip and a symptom,
reason step by step:
first check whether the symptom allows travel,
then look up the
company doctor's consultation hours and booking
rules, then summarize.
```



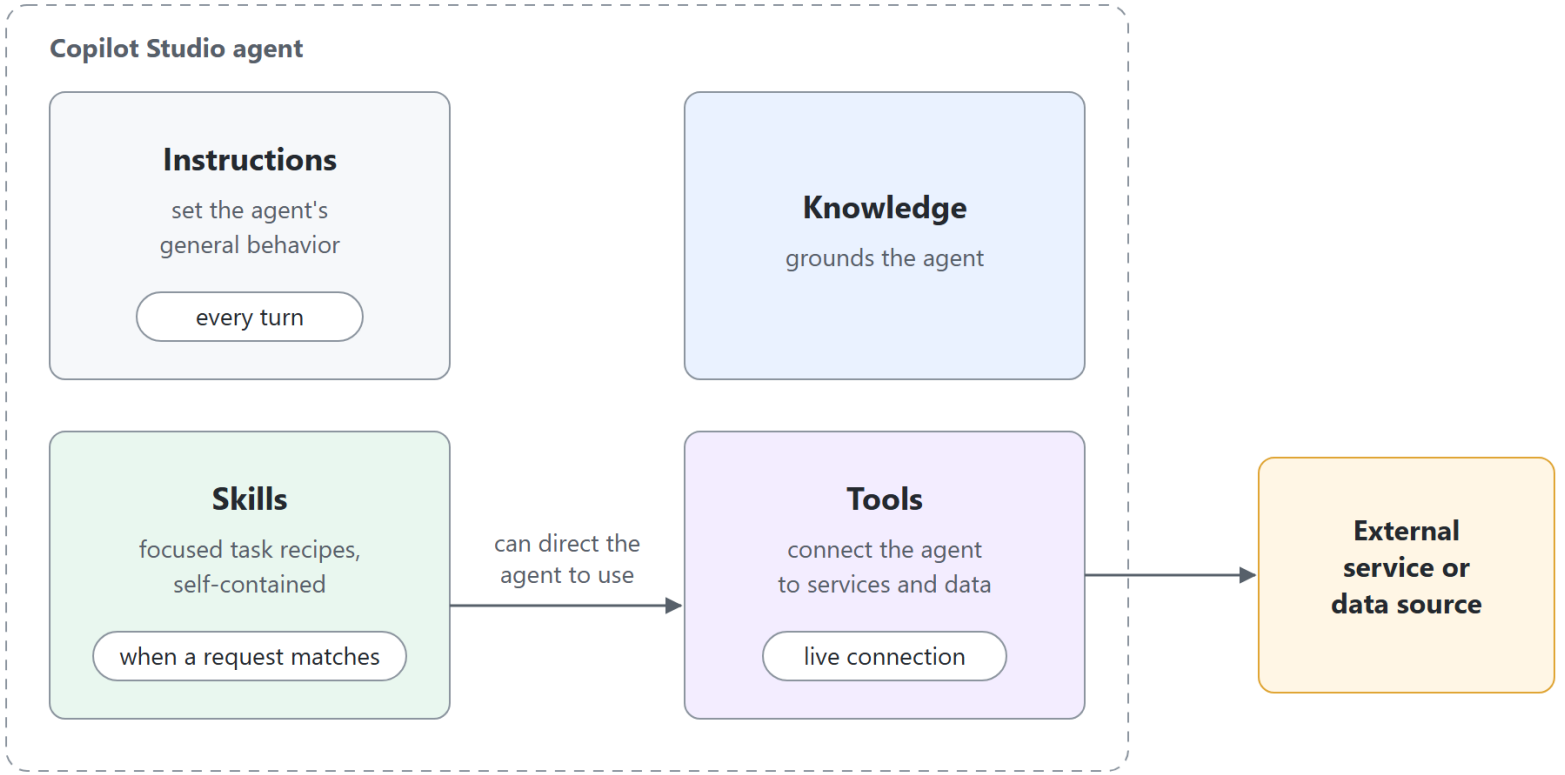
03

Agent Skills in Copilot Studio

SKILL.md recipes that load when a request matches

What a skill is

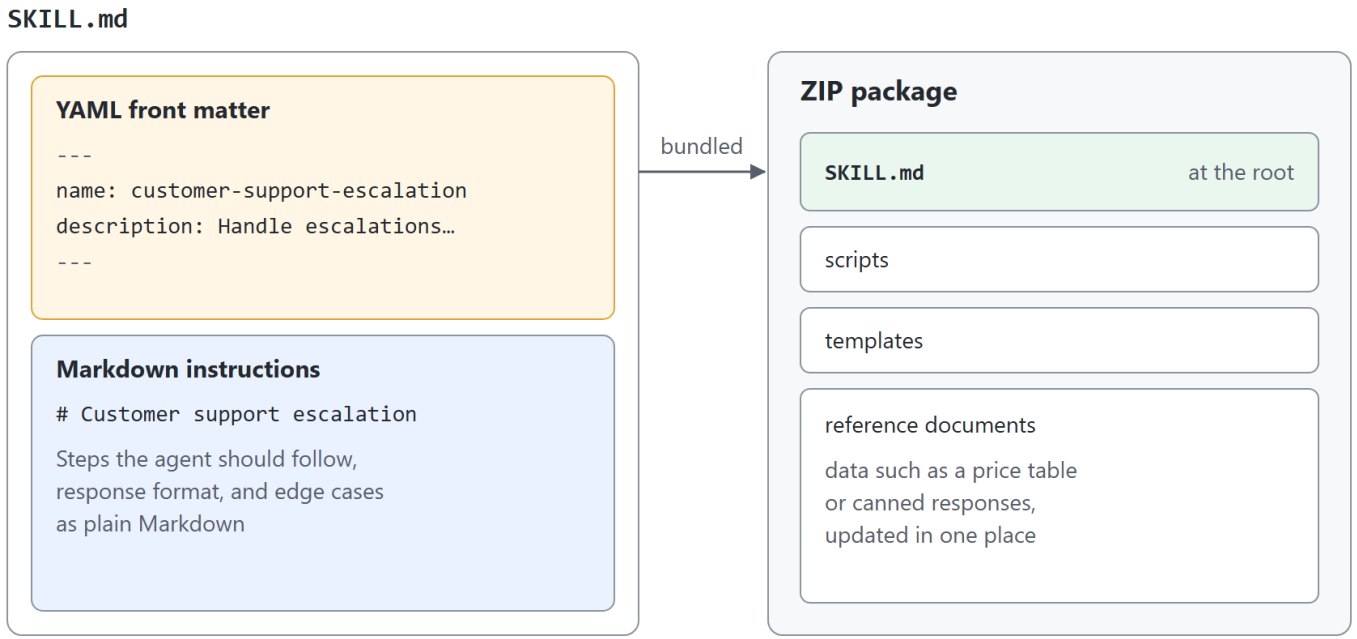
A playbook for one recurring task, next to instructions, knowledge and tools



A **skill** packages the recipe for one task: a name, a description and Markdown instructions the agent loads only when a request needs them

What a skill file holds

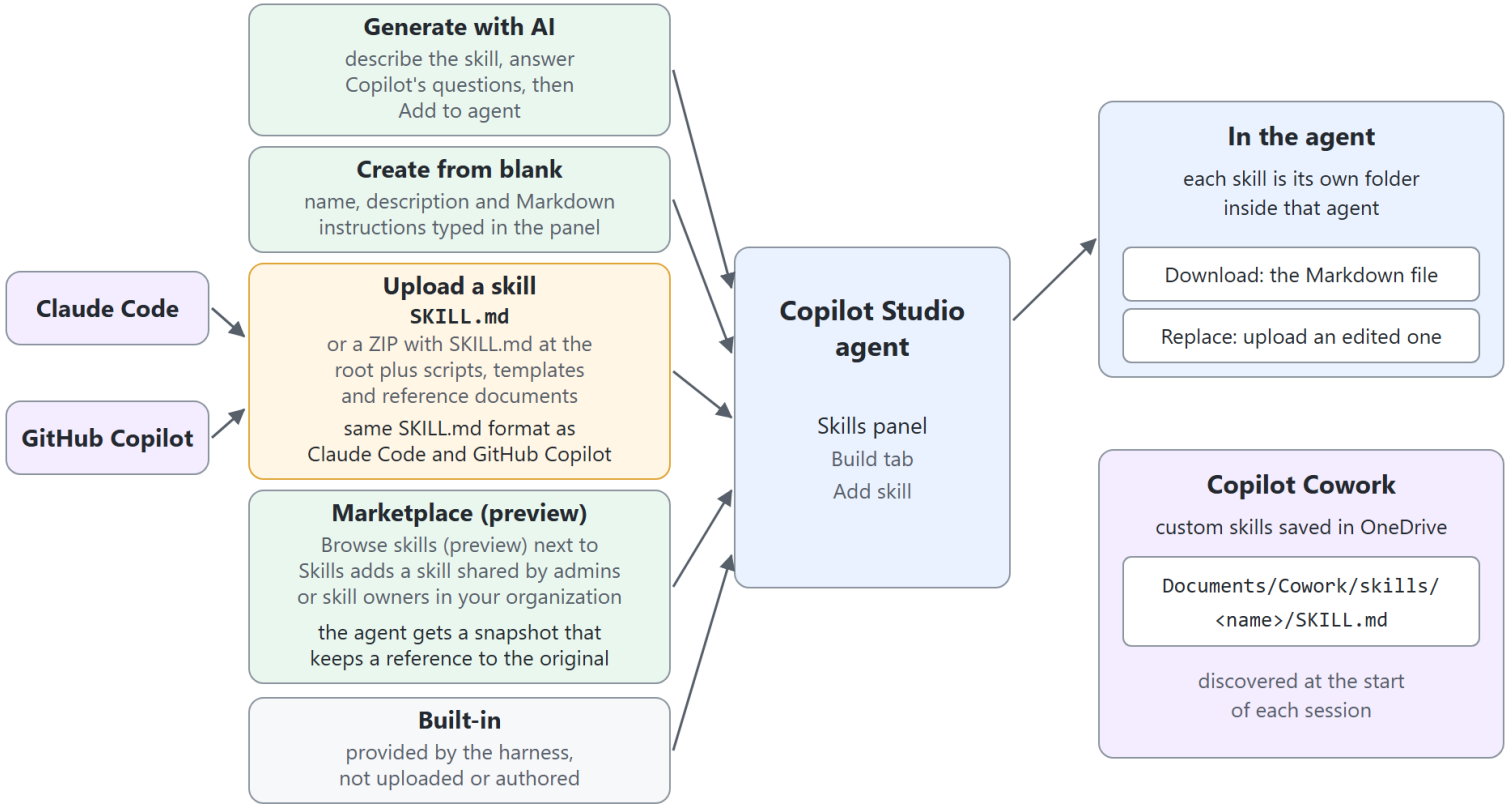
A SKILL.md on its own, or in a ZIP with its supporting files



SKILL.md: YAML front matter with the name and description, then Markdown instructions; a ZIP adds scripts, templates and reference files, with SKILL.md at the root

Where skills come from and where they are saved

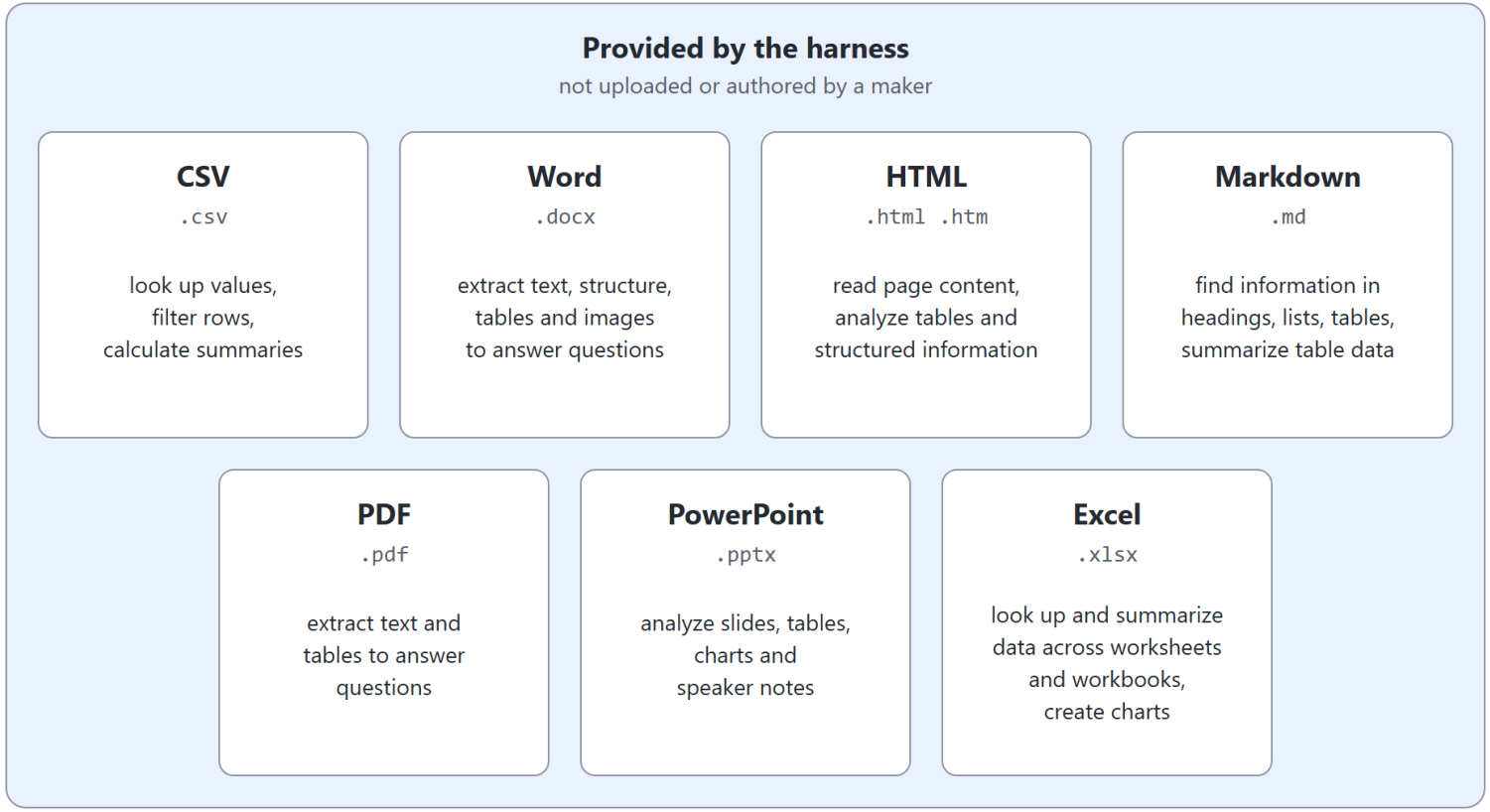
Into the agent's Skills panel, or into OneDrive for Copilot Cowork



Generate with AI, **Create from blank**, **Upload a skill** or the marketplace saves it in the agent; Copilot Cowork saves custom skills in OneDrive under Documents/Cowork/skills

Built-in skills

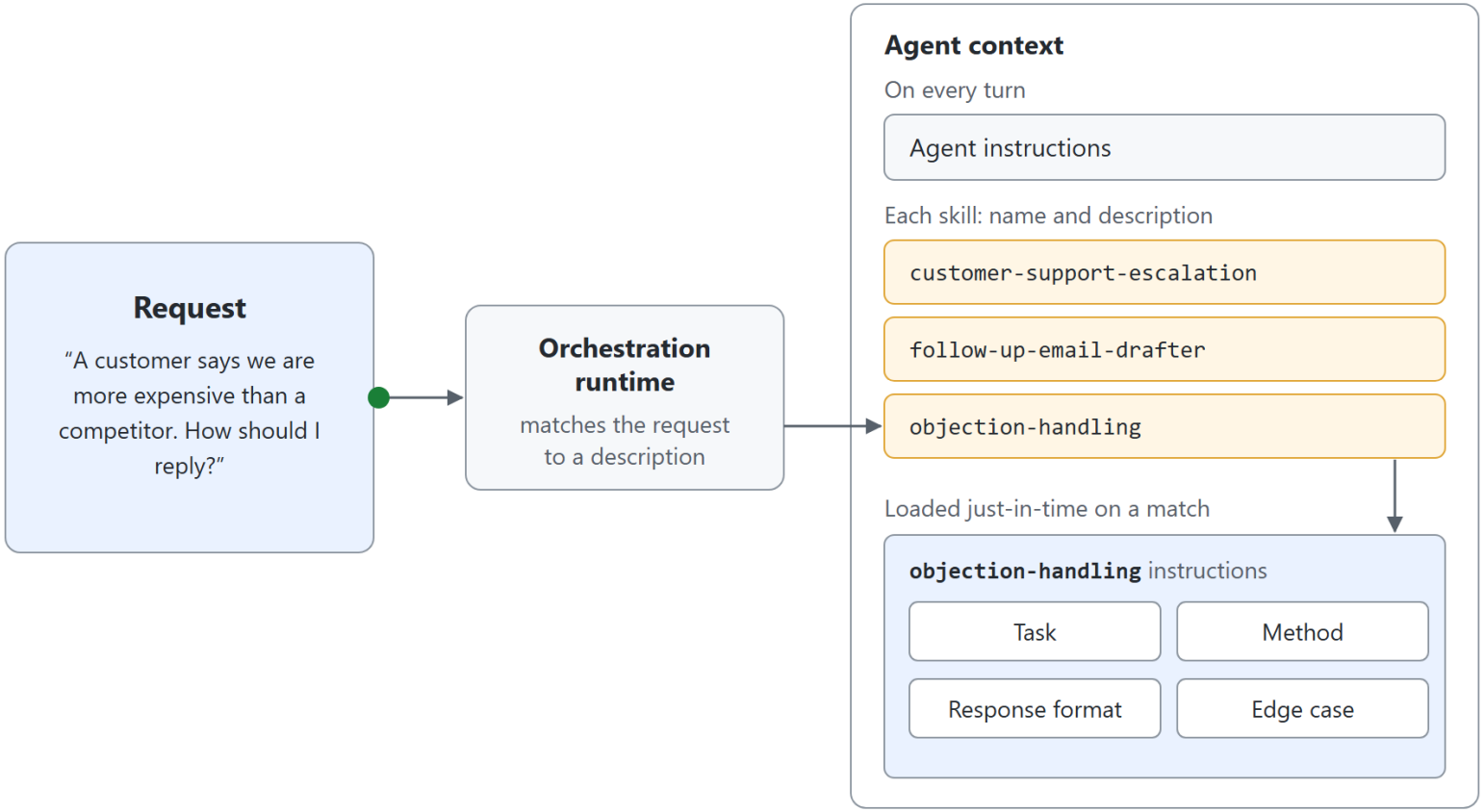
Document and data skills the harness brings along



Seven **built-in skills** read CSV, Word, HTML, Markdown, PDF, PowerPoint and Excel files; the harness provides them, nobody uploads or authors them

How a skill loads on demand

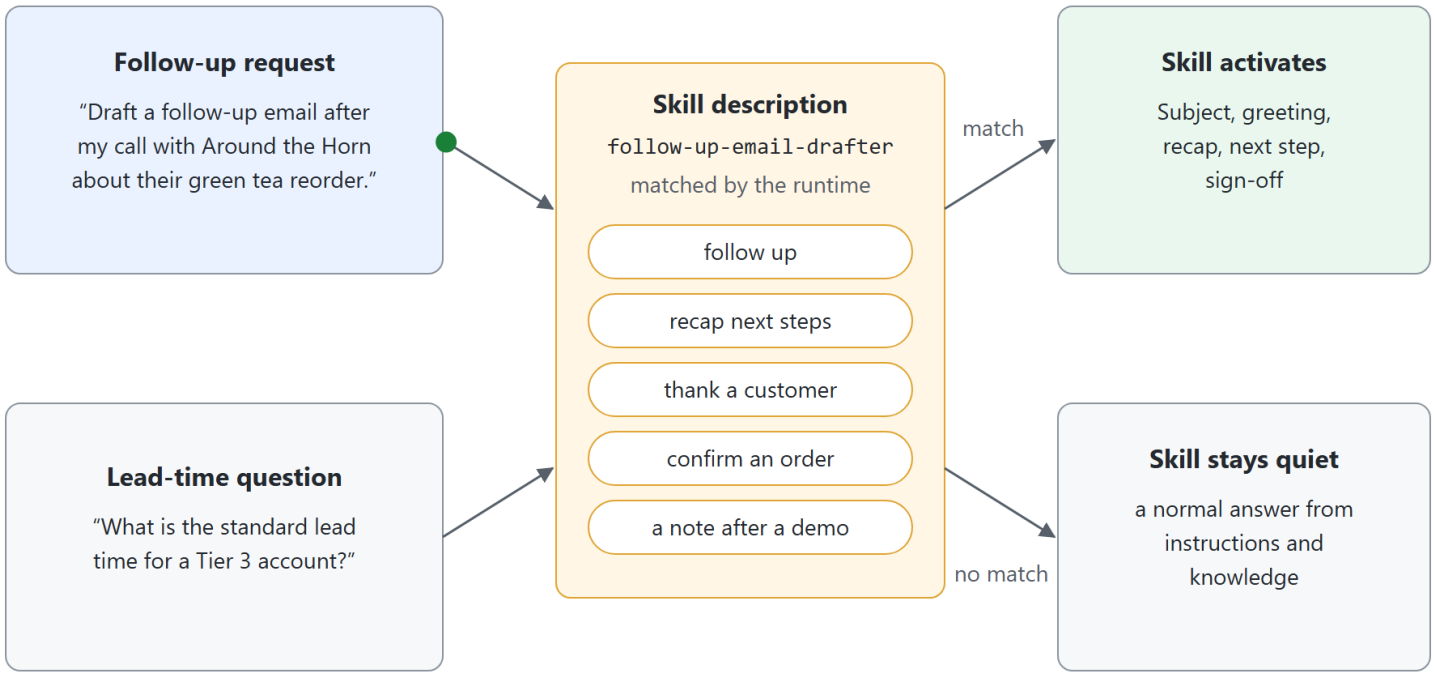
Only names and descriptions stay in context until a request matches



The runtime matches a request to a skill's **description** alone, then loads that skill's full instructions just-in-time

Testing a skill in Preview

The description decides when the skill fires



A request in the description's words activates the skill; a near-miss question must stay quiet and get its answer from instructions and knowledge

04

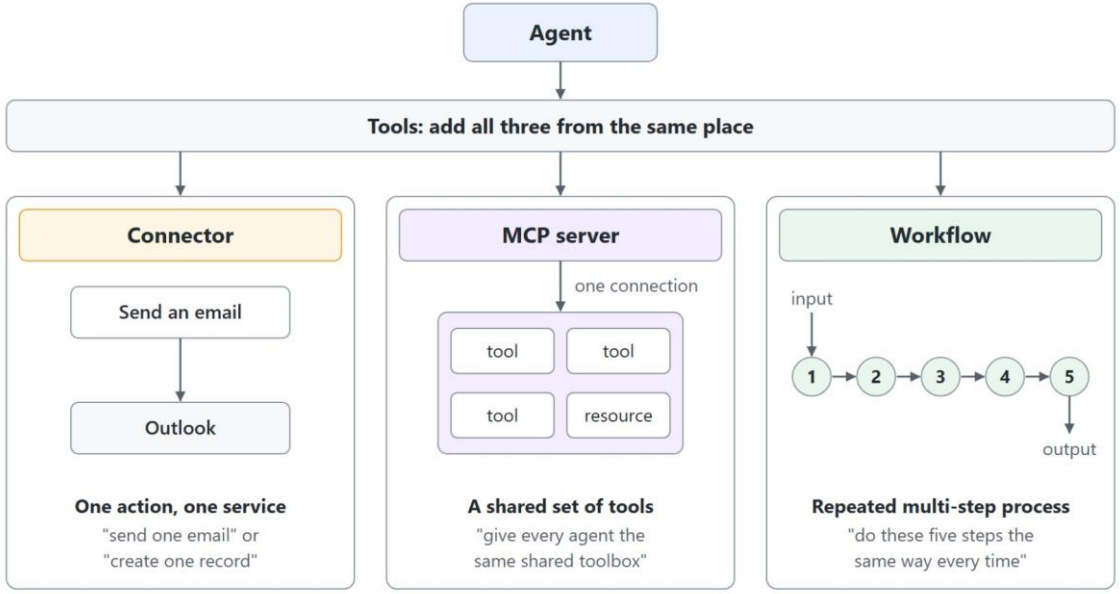
Tools, MCP, and the New Workflows Designer

Connectors, MCP servers and workflows with agent nodes

Connectors, MCP servers and workflows

Which tool fits which job

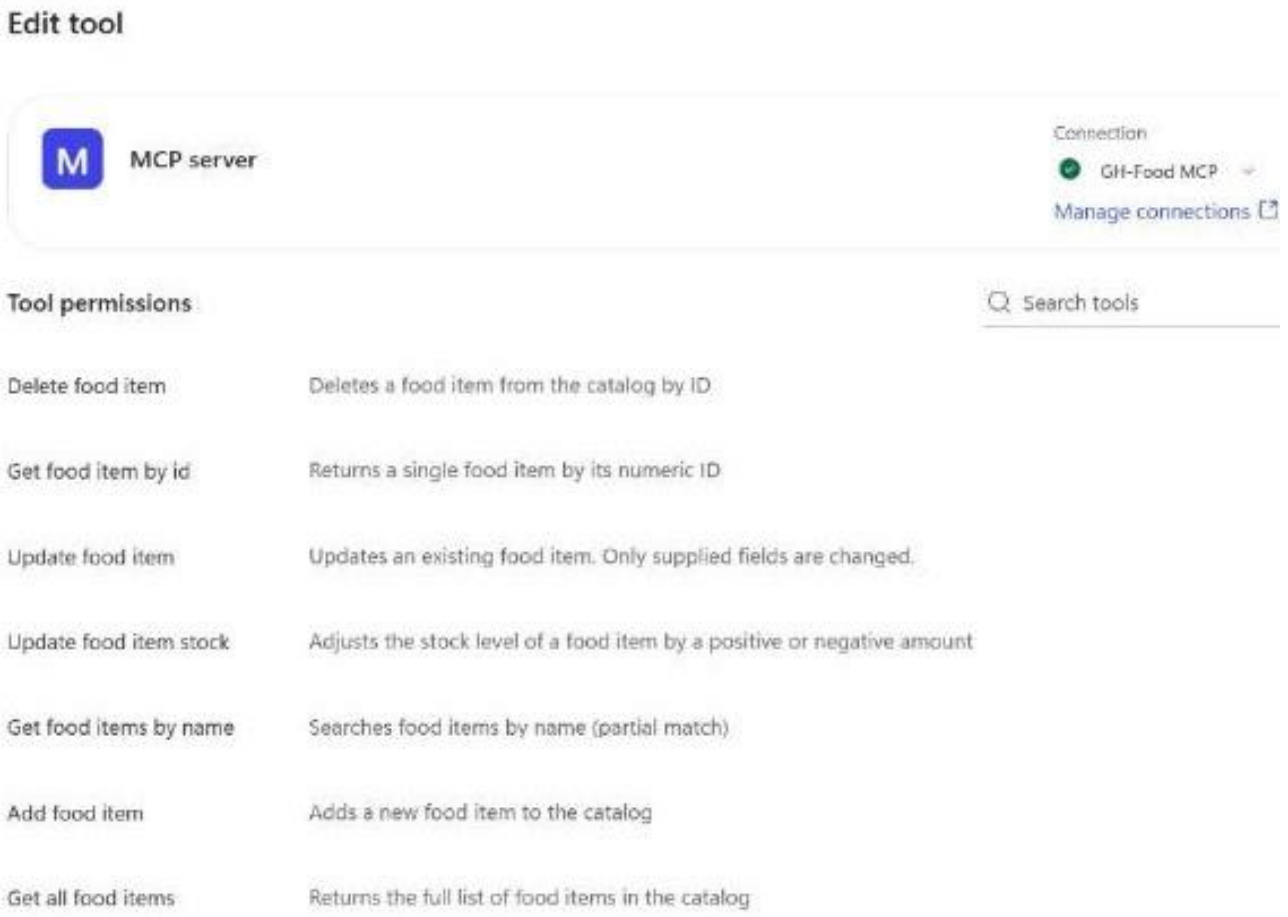
- **Connector**: one action against one service, such as sending one email
- **MCP server**: one connection to a shared set of tools; its description tells the orchestrator when to call it
- **Workflow**: a repeated multi-step process, tested node by node and versioned in the new designer
- **Agent node**: a workflow step that hands a task to your agent and passes the result on



Adding an MCP server

One connection, every tool the server exposes

- **Add a tool**, MCP tab for a prebuilt server, or **Model Context Protocol** for your own
- Server name, **description**, URL and authentication: None, API key or OAuth 2.0
- **Edit tool** lists every tool the server exposes under **Tool permissions**
- The orchestrator calls the server on its own when a question matches the description



A workflow with an agent node

Fixed steps around one call to your agent



- Trigger, **Agent** node and **Respond to the agent** ran in 17 seconds; the agent node called the published GH-Food Buddy

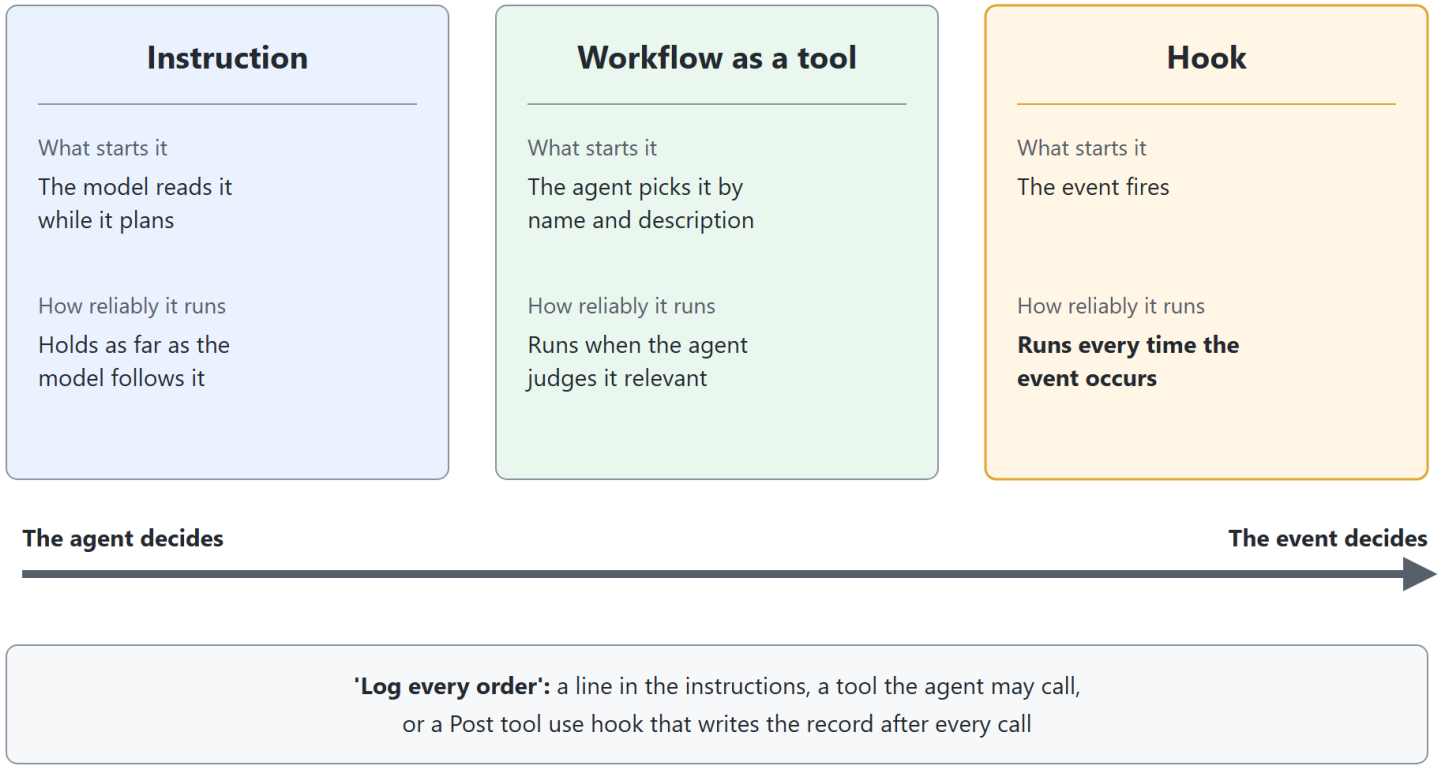
05

Hooks: Steps That Run Every Time

Workflows bound to session, prompt, tool and error events

Who decides that a step runs

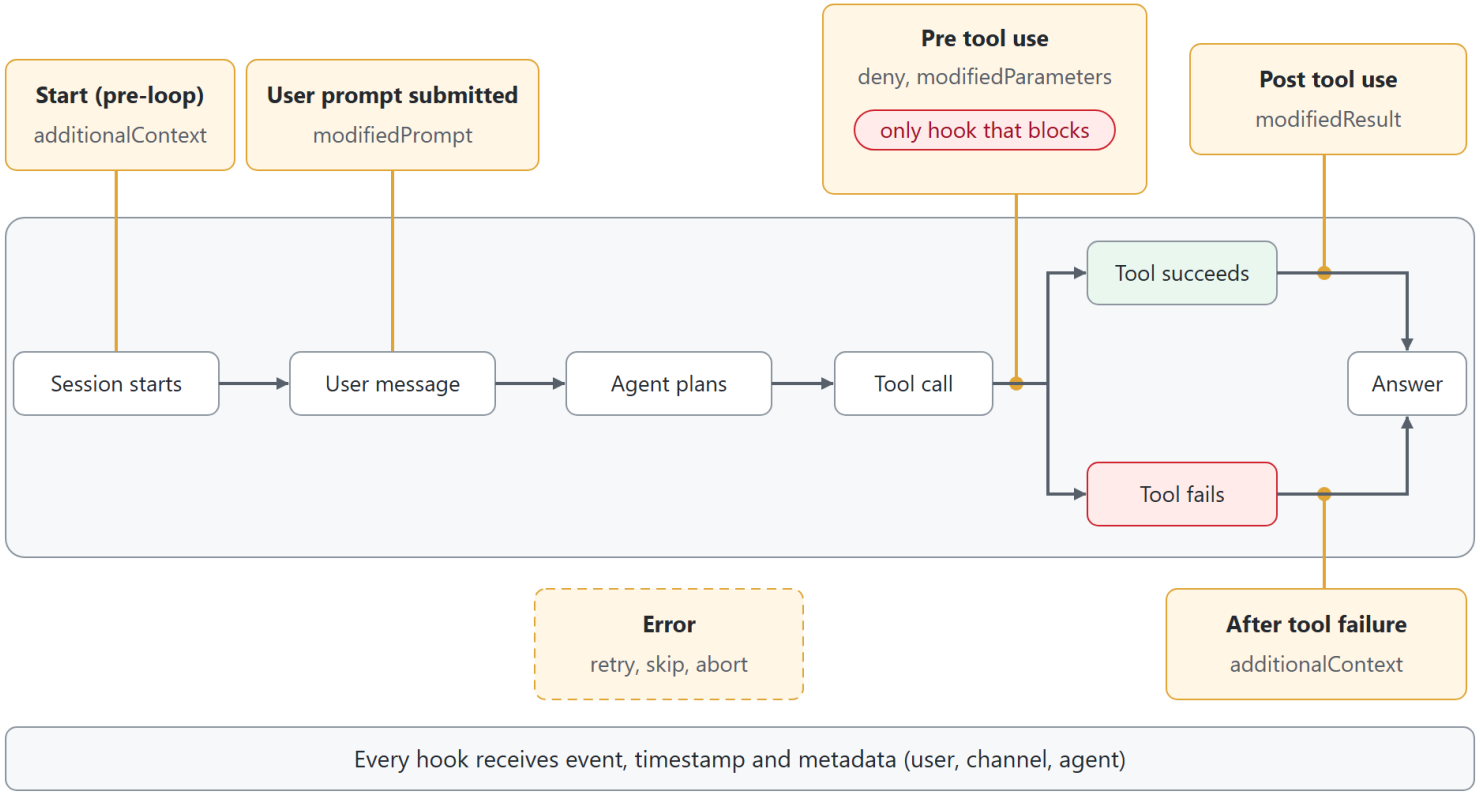
Instructions guide the model, a tool waits to be chosen, a hook runs on its event



A rule that must hold on every session or call becomes a hook

Hooks at six points of a run

Each event runs a workflow that can change what happens next

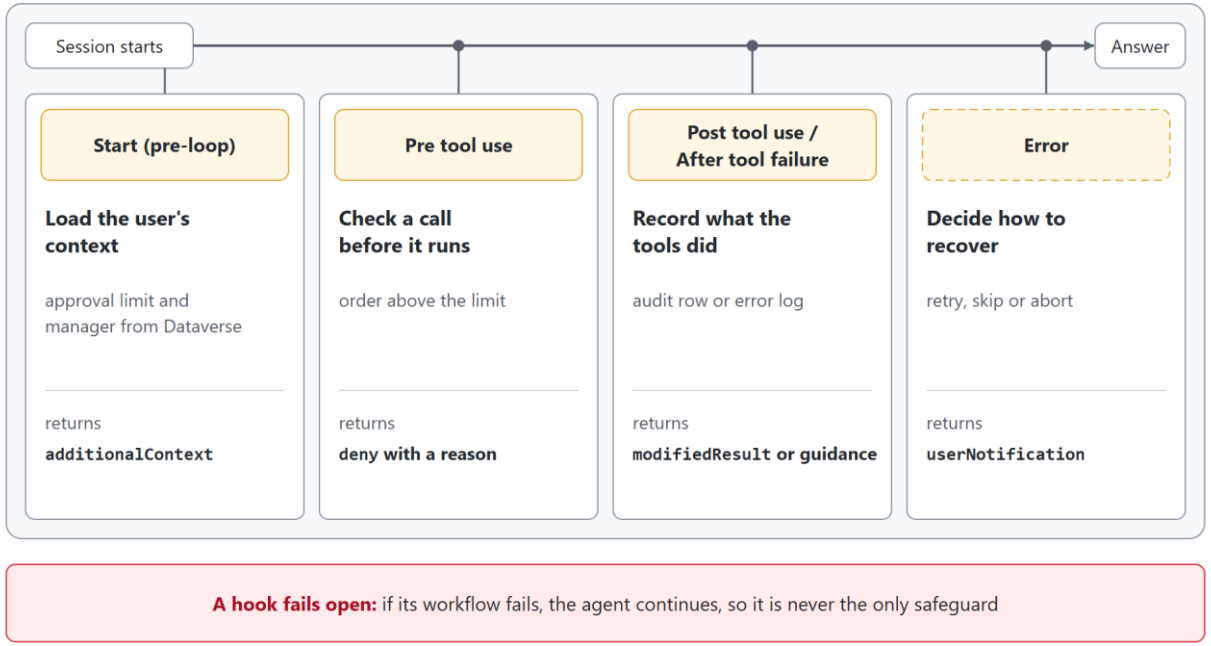


Each hook runs a workflow; only **Pre tool use** can block a call

Four patterns worth building

Load context, check a call, record the result, recover from an error

- One pattern per stage of the run, each a short published workflow



06

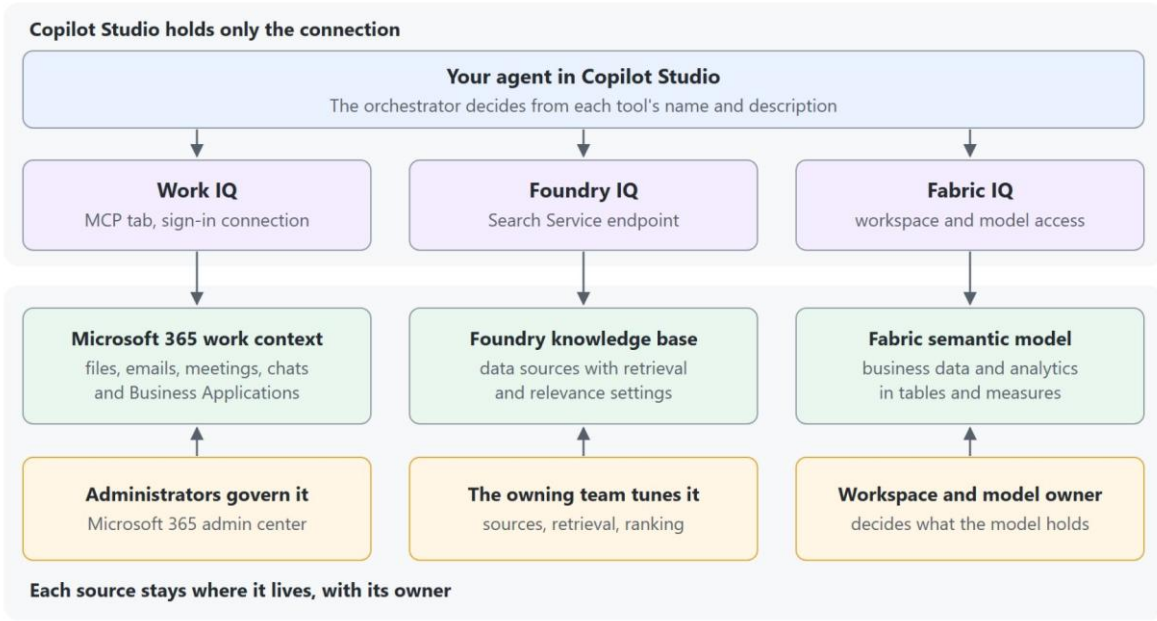
Grounding Agents with Work IQ, Foundry IQ, and Fabric IQ

Sources that stay where they live, proven in the trace

Grounding with Work IQ, Foundry IQ and Fabric IQ

The agent reads each source where it lives

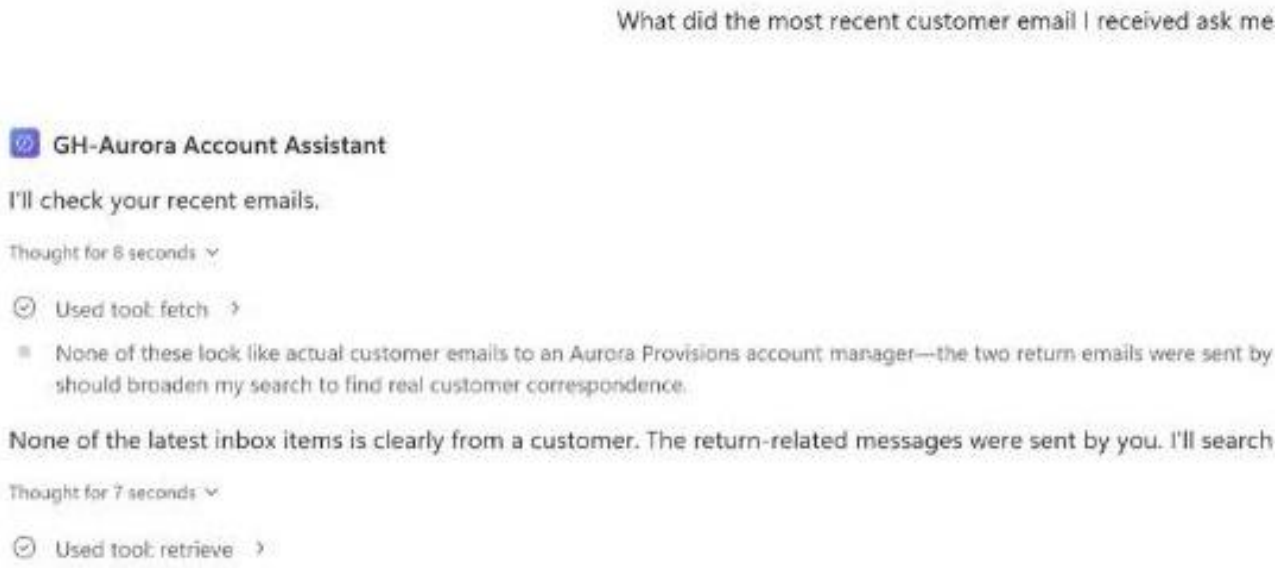
- **Work IQ** reads your Microsoft 365 email, meetings, chats and files
- **Foundry IQ** answers over a knowledge base a team tunes in Microsoft Foundry
- **Fabric IQ** answers data questions from a semantic model in Microsoft Fabric
- Copilot Studio holds only the **connection**; each source stays with its owner



Proving grounding in the activity trace

The same question, asked again after Work IQ is added

- Add **Work IQ (Preview)** from the MCP tab of **Add a tool**, not the legacy Mail or Teams tools
- Ask the **same question** before and after: only the tool changed
- The trace shows **Used tool: retrieve** and **fetch**; expand a line for its input and output
- **Debug mode** must be on to expand the lines



Live sources and uploaded files

Why grounding matters for information that moves

- An **uploaded file** is an indexed copy: the agent knows what you uploaded last
- An **IQ connection** reads the source where it lives when the question arrives
- That matters for information that moves, such as a **customer email from this morning**
- You change a file by uploading a new version; an IQ source is tuned by its **owner**



07

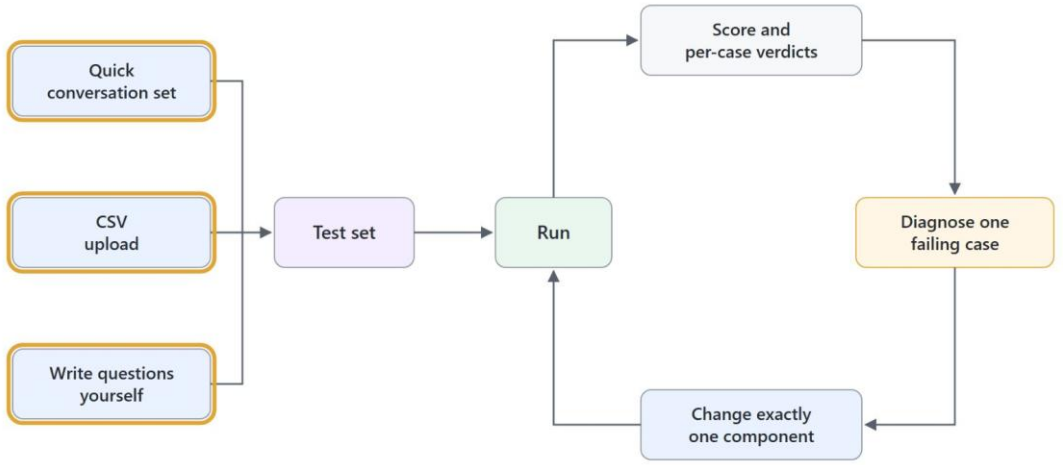
Test and Evaluate Copilot Studio Agents

Test sets, General quality and a failing case to its fix

Evaluating agents with test sets

Fill a set, run it, change one thing, run it again

- **Test set**: conversations of user questions with optional expected responses, filled by Quick conversation set, CSV upload or by hand
- **General quality**: the one test method on the Evaluate tab; it judges relevance and completeness and does not compare to the expected response
- **User profile**: the identity a run executes as, matched to the access real users have
- **Invalid and Error**: about the run or the case, so clear them before reading Pass and Fail
- **One change per run**: baseline, one change, re-run the same set, compare the cases that flipped



A test set worth trusting

Weight the cases toward where the agent is likely to fail

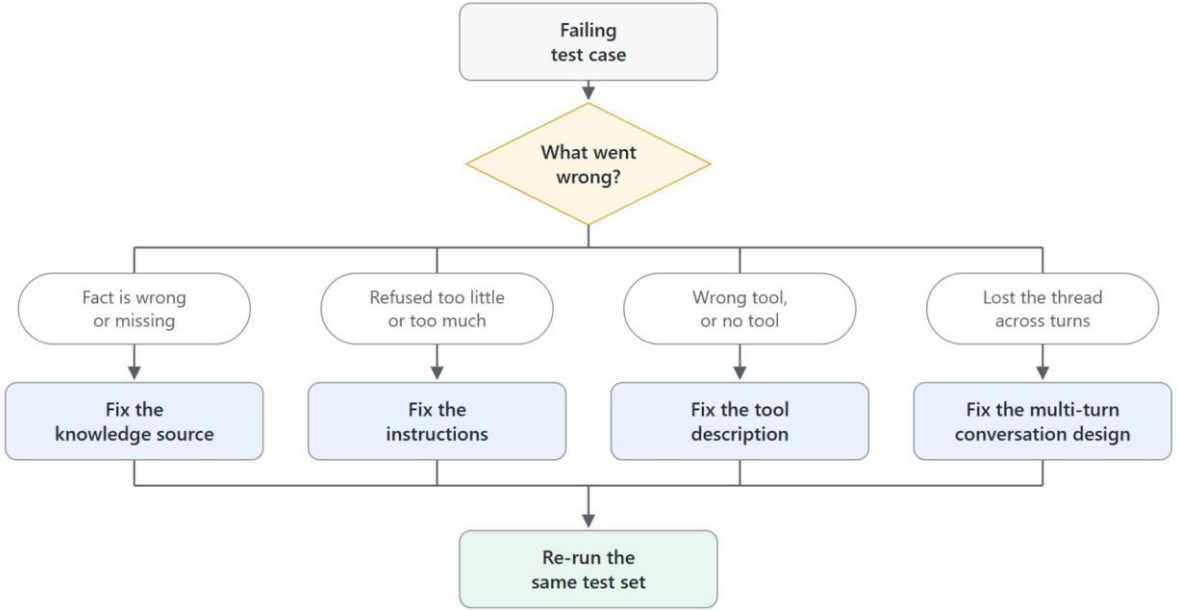
- **Grounded facts**: the agent reports something from its knowledge correctly
- **Guardrails**: it refuses what it should, such as a binding price quote
- **Off-topic**: it declines instead of answering from general model knowledge
- **Multi-turn**: it holds context and asks for the missing detail



From a failing case to a fix

What went wrong names the component to change

- **Wrong or missing fact**: fix the knowledge source
- **Refused too little or too much**: fix the instructions
- **Wrong tool or no tool**: fix the tool description
- **Lost the thread across turns**: fix the multi-turn design, then re-run the same set



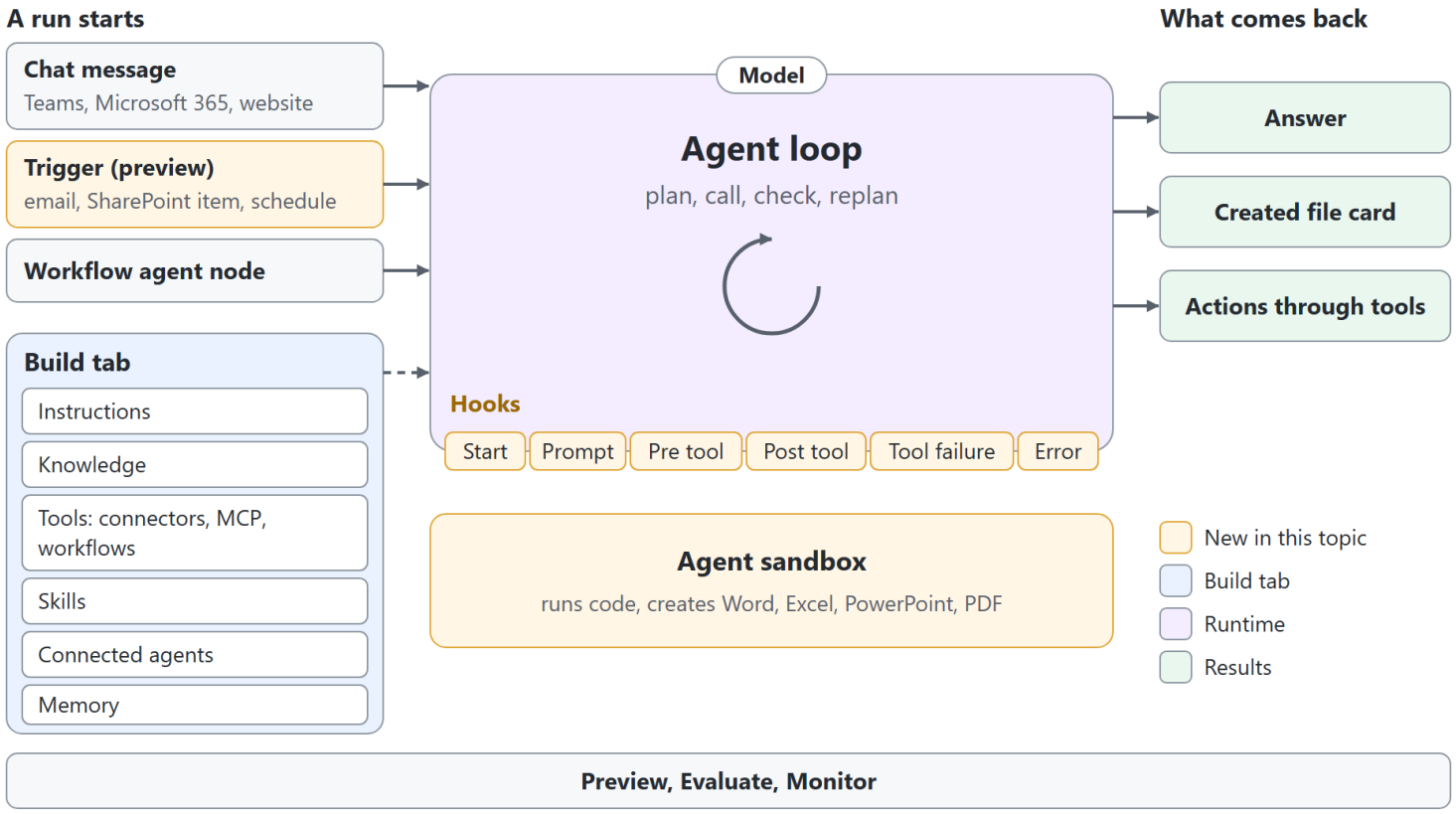
08

Migrate Standard Agents to Microsoft Copilot Harness Agents

When to move, three routes, topics as workflows and hooks

The full harness at a glance

Everything around the model, and the three parts this topic adds



Amber marks the three parts this topic adds: hooks, triggers and the sandbox

Migrate for a named gain

A working Standard agent doesn't have to move

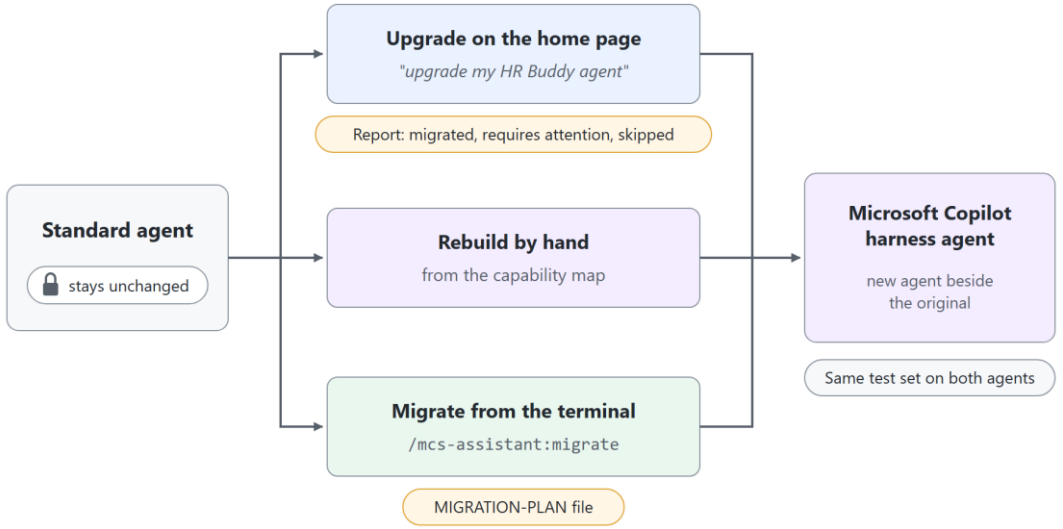
- Move for a **named gain**: quality, long runs, a reasoning model, skills or memory
- **Stay Standard** for fixed intake, rich cards and channels the new agents lack
- Treat it as **re-architecture**: tasks, outcomes and test set first
- **Done** when the core journeys pass the same test set



Three routes to a migrated agent

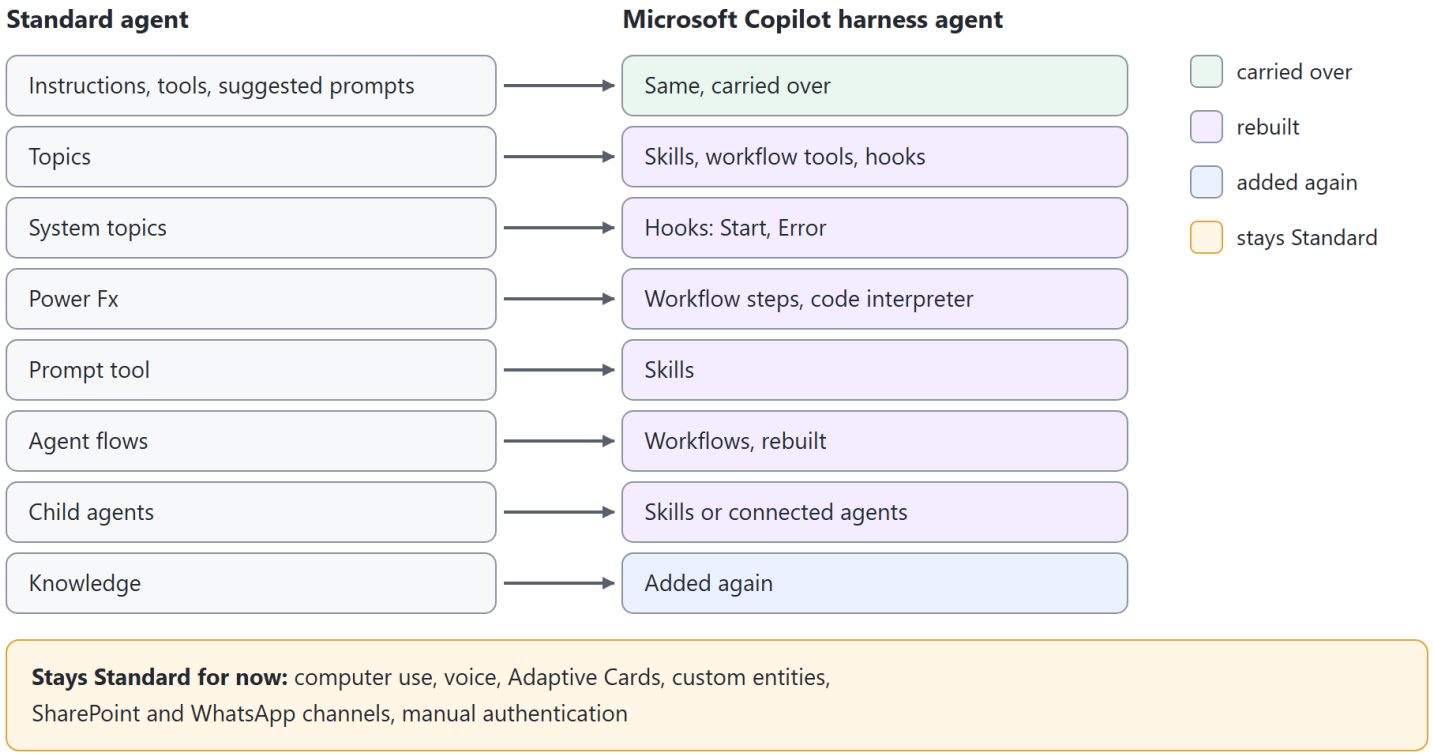
Each one leaves the Standard agent untouched

- **Upgrade**: type upgrade my <agent> agent on the home page
- Carries **instructions, tools and suggested prompts**, then reports
- **Not migrated**: knowledge sources, Power Fx and child agents
- **Rebuild by hand** from the map, or **migrate from the terminal**



Where each Standard part lands

Carried over, rebuilt, or staying on Standard agents

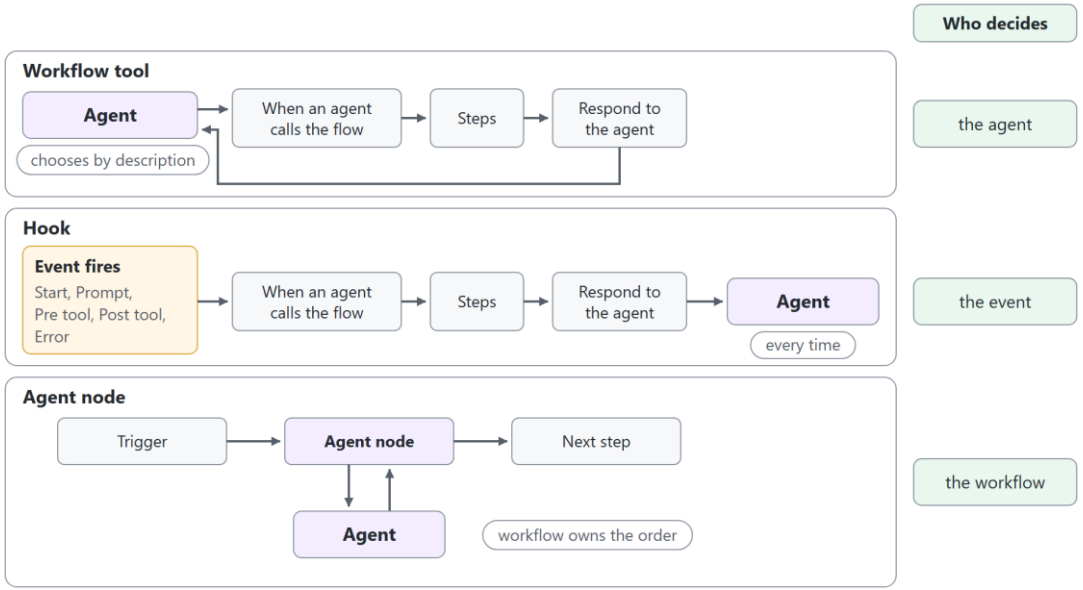


Each topic becomes a **skill**, a **workflow tool** or a **hook**, by purpose

Fixed steps without topics

A workflow tool, a hook, or a workflow that calls the agent

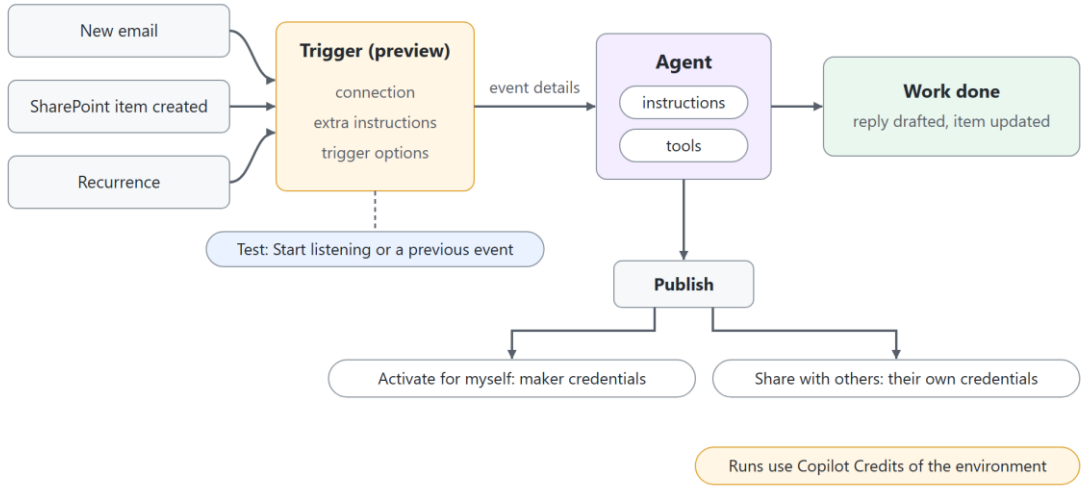
- **Workflow tool**: fixed steps, called when the agent picks it
- **Hook**: its workflow runs **every time** the event fires
- Only **Pre tool use** can block a call, with permissionDecision
- **Agent node**: the workflow owns the order, the agent does one step



Triggers start the agent on an event

An email, a SharePoint item or a schedule, with no one chatting

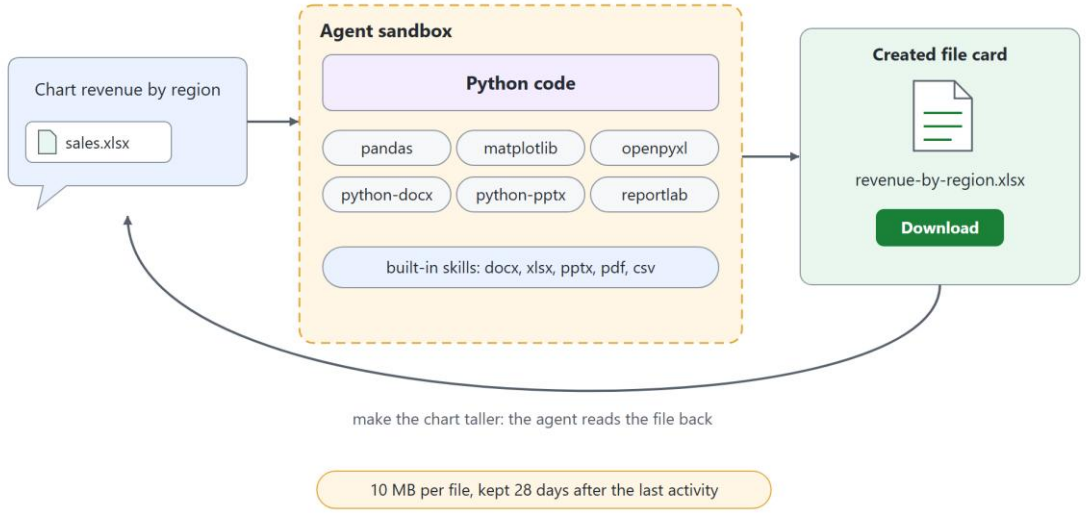
- The **trigger** sends event details, instructions decide the work
- Set the **connection**, extra instructions and trigger options
- Test with **Start listening** or a previous trigger event
- **Publish**, then activate for yourself or share with others



The sandbox runs code and builds files

Word, Excel, PowerPoint and PDF without adding a tool

- Each task runs in a **secure sandbox** governed by Copilot Studio
- Python packages build **Word, Excel, PowerPoint and PDF** files
- A **created file card** offers the file for download
- Each file stays **28 days**, up to **10 MB**



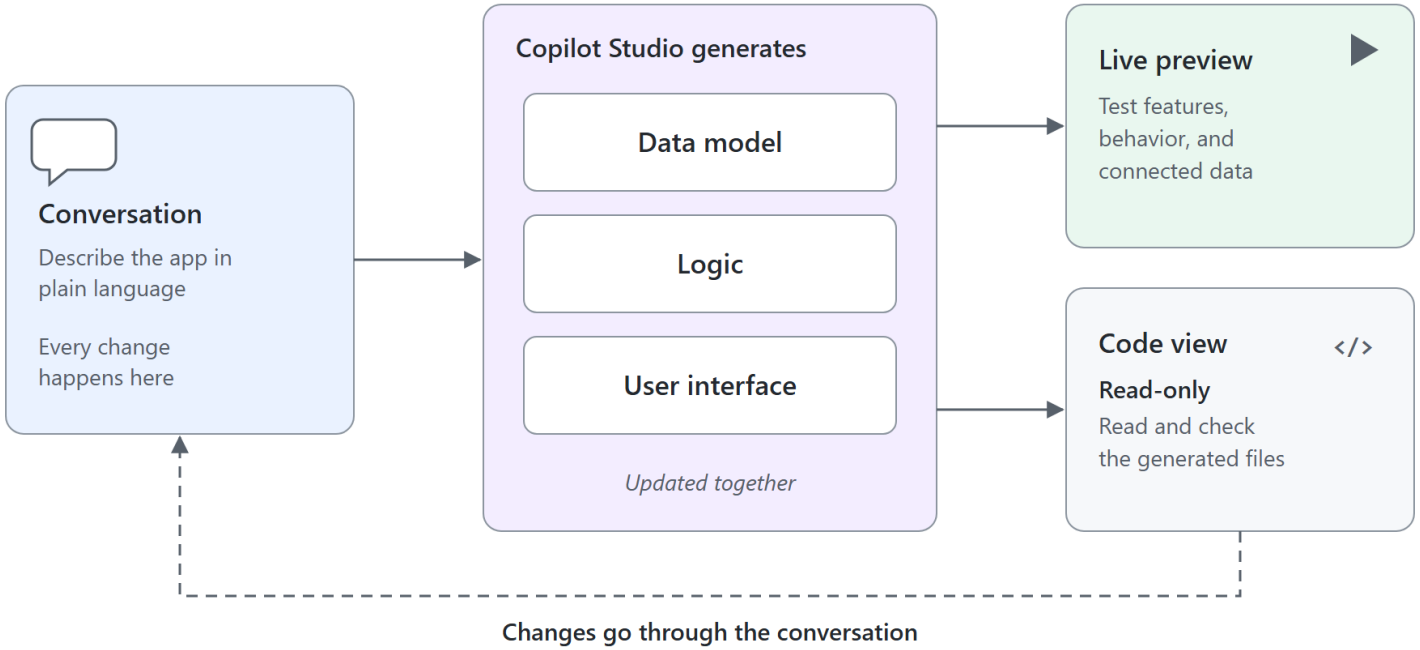
09

Building Apps in Copilot Studio

Describe, refine by conversation, publish and share

What an app is

A web app you describe, and Copilot Studio builds



An **app** is an interactive web experience: Copilot Studio generates its data model, logic and user interface from your description, and every change goes through the conversation

How these apps compare

The whole app from a description, next to Cowork and Power Apps

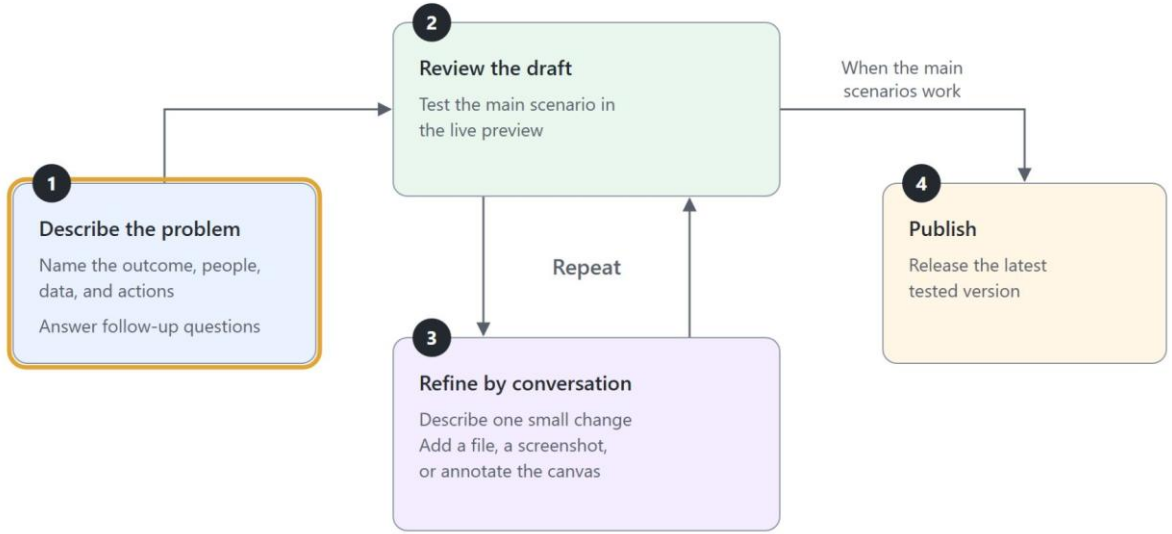
App in Copilot Studio preview	App skill in Cowork Copilot Cowork, Frontier	Canvas app Power Apps	Model-driven app Power Apps
How you build it Describe the need in plain language Refine it in the conversation Preview as you build	**How you build it** Describe the app in a Cowork conversation No coding required	**How you build it** Start from a blank canvas Place controls and write Power Fx	**How you build it** Start from the data model in Dataverse
What you get Data model, logic and interface, generated App, data and connections in one place Data access with the user's own permissions Publish and share, Microsoft-hosted	**What you get** A lightweight, interactive app Refine, open, publish and share	**What you get** A screen designed control by control	**What you get** Forms and views built on that data Layout from the components you add

An app in Copilot Studio is generated whole from a description; a canvas app is designed control by control, a model-driven app starts from Dataverse tables, and Cowork's App skill builds a lighter app in a conversation

Building apps

Describe the app, then review and refine it until the main scenarios work

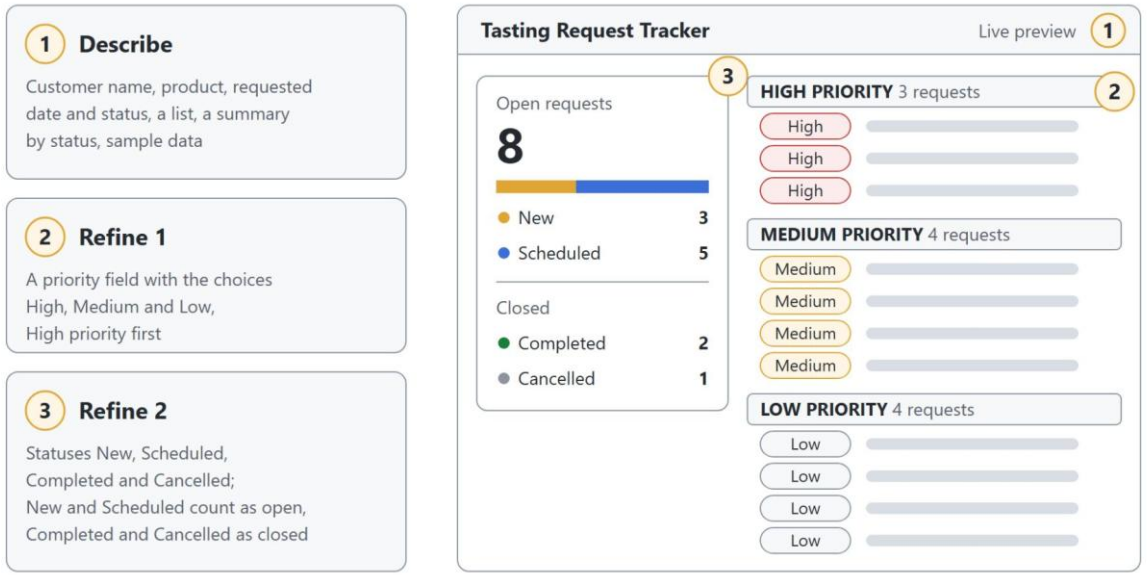
- **Describe**: name the outcome, people, data and actions; Copilot Studio generates the data model, logic and interface
- **Refine**: one small change per message, checked in the live preview; the code view is read-only
- **Data access**: granted on the SharePoint list or Dataverse table, separately from sharing the app
- **Copilot Credits**: billing starts as soon as you start building



The Tasting Request Tracker

An app refined by conversation, checked in the live preview

- **Describe**: one message named the people, the data and the actions; the first draft took about five minutes
- **Refine 1**: a priority field, with High requests listed first
- **Refine 2**: four statuses, with New and Scheduled counted as open
- **Check** each change in the preview before sending the next one



10

Customer Conversation Agents: Topic Skills, Sub-Agents, and Escalation

Sub-agents, two skill kinds, escalation

Three harnesses, three kinds of agent

Microsoft Copilot, Customer Conversation and standard

- The harness is picked at creation and decides how the agent works

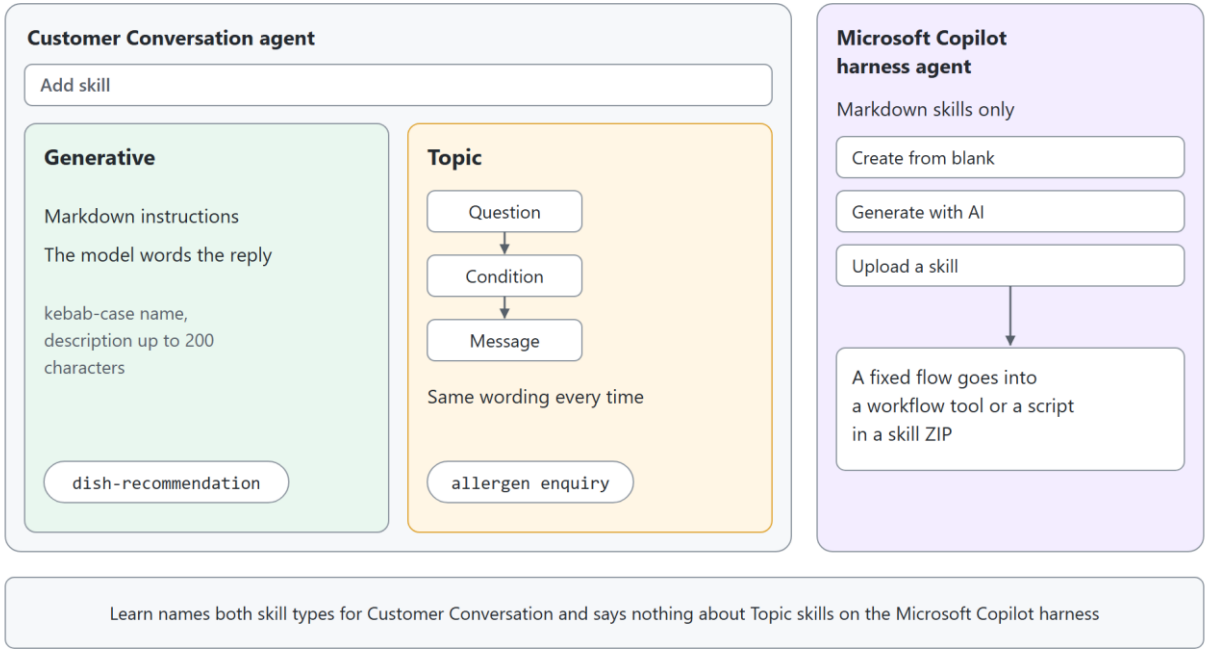
	Microsoft Copilot harness	Customer Conversation harness	Standard harness
Best for	Complex multi-step processes	Customers over voice and chat	Rule-based conversations
How it works	Reasons through a goal	Primary agent routes to sub-agents	Follows your topics
When it goes wrong	Retries, finds another path	Escalates to a person	Follows the paths you built
Skills	Markdown skills	Generative and Topic skills	Topics
Billing	Copilot Credits	Copilot Credits	Standard licensing

Tell them apart: only a Customer Conversation agent has a Deploy tab

Generative skills and Topic skills

The model reasons, or your flow decides

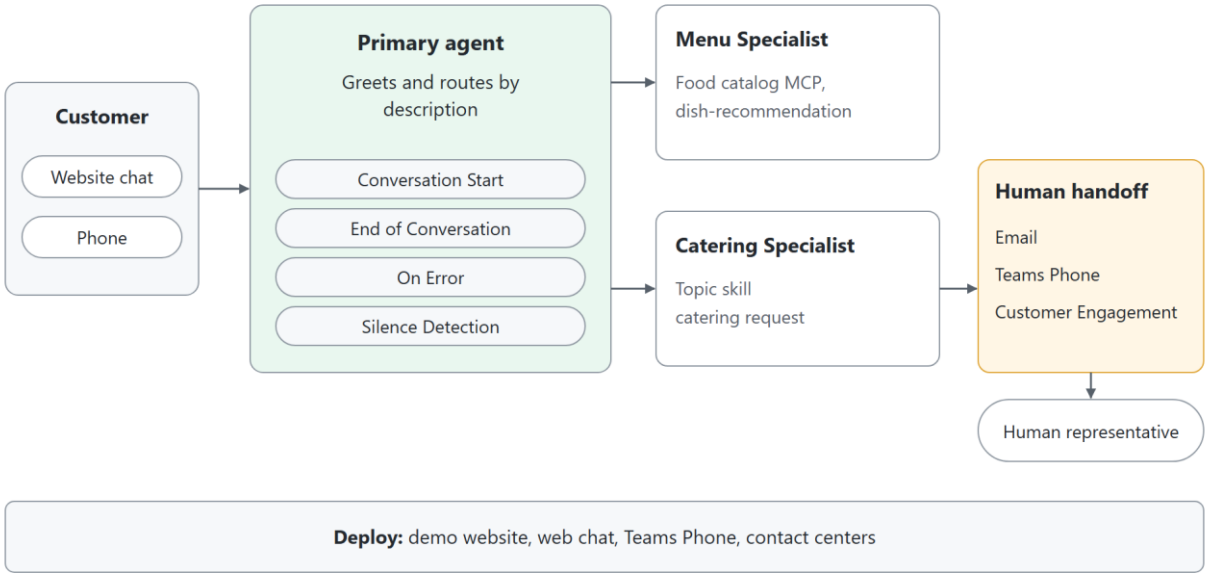
- A topic's description must claim the data it collects



Sub-agents and a way to a person

Route by description, escalate by email, phone or live agent

- Build sub-agents like departments, each with its own escalation



Takeaways

What to carry into publishing and maintaining

- Decide the **kind of agent** before you build: there is no conversion
- Read the **step list** and **History** to see how the orchestrator planned
- A skill's **description** and an MCP server's **description** decide when they are used
- Prove grounding in the **activity trace**, not in the wording of the reply
- Run the **same test set** before and after every change

